

On aspectual coercion in *before*-/*after*-clauses: Evidence from processing

Johanna Alstott
jalstott@mit.edu

Athulya Aravind
athulya.aravind@yale.edu

Abstract. Certain *before*-clauses and *after*-clauses are *prima facie* ambiguous between a strong reading and a weak reading, and theories diverge on how the readings obtain. Some theories take the LF of *before*- and *after*- clauses to be unambiguously weak, with the strong reading falling out as a subcase (as it entails the weak reading). Other theories, such as Rett (Sinn und Bedeutung 24, 201–218, 2020), take the LFs for the two readings to be distinct, critically, with the weak reading being the result of a coercion process. Under standard psycholinguistic assumptions, coercion incurs a processing cost, and hence, we argue, Rett’s (2020) theory makes processing predictions that are not shared by theories which posit unambiguously weak LFs. We test these predictions via four self-paced reading experiments and find moderate evidence that they are borne out. Our results constrain theories of *before* and *after*, contribute to a typology of aspectual coercion mechanisms, and have implications for how we should think about the licensing conditions of coercion.

1 Introduction

The temporal connectives *before* and *after* exhibit two truth-conditional asymmetries. First, as originally noted by Anscombe (1964), sentences with atelic predicates in *after*-clauses (1a) are *prima facie* ambiguous (i.e. they can be thought of as true in two types of scenarios), whereas parallel sentences with *before* are not. To see this, suppose that Emma and Hector are at a party and Hector was happy from 6pm–8pm; now, consider the scenarios in Figure 1 for when Emma left the party (Scenario A, 5:30pm; Scenario B, 7pm; Scenario C, 8:30pm). (1a) is true if Emma left after Hector stopped being happy (Scenario C) or if she left after Hector started being happy (Scenario B). By contrast, (1b) is only true if Emma left before Hector started being happy (Scenario A). (1b) lacks a “before-finish” reading: if Emma left before Hector stopped being happy but not before he started (Scenario B), (1b) is false.

- (1) a. Emma left after Hector was happy.
(✓after-start, ✓after-finish)

- b. Emma left before Hector was happy.
 (✓before-start, ✗before-finish)

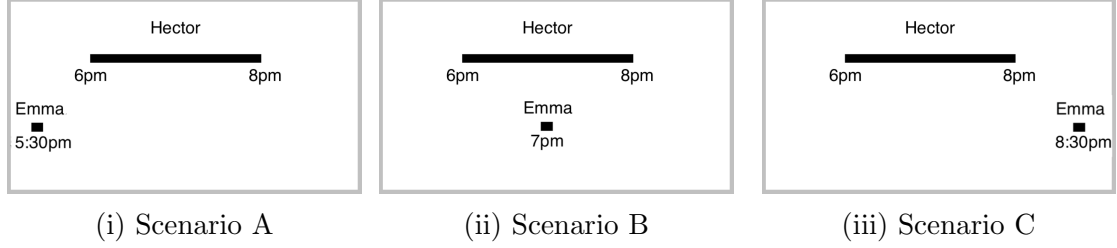


Figure 1: Supplemental timelines for (1).

Second, as originally noted by Heinämäki (1974), sentences with accomplishments in *before*-clauses (2a) are *prima facie* ambiguous, whereas parallel sentences with *after* (2b) are not. To see this, suppose that Jackson climbed a mountain in twelve hours (9am-9pm) and that there was an all-day watch party for the event that started at 6am. Now, consider three scenarios for when Emma left the watch party (Scenario A, 7am; Scenario B, 3pm; Scenario C, 11pm). (2a) is true if Emma left before Jackson started climbing the mountain (Scenario A) or if she left before he finished (Scenario B). By contrast, (2b) is only true if Emma left after Jackson finished (Scenario C). If Emma left after he started but not after he finished (Scenario B), (2b) is false.

- (2) a. Emma left before Jackson climbed the mountain.
 (✓before-start, ✓before-finish)
 b. Emma left after Jackson climbed the mountain.
 (✗after-start, ✓after-finish)

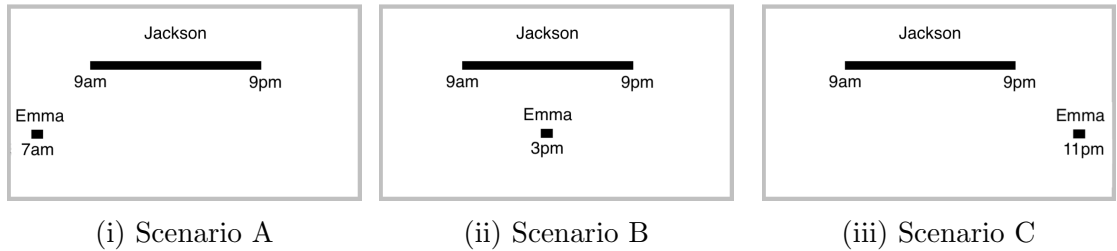


Figure 2: Supplemental timelines for (2).

Historically, the asymmetry in (1) has received more attention than the facts about accomplishments; two types of theories of (1) have been proposed, which we

will call under-specification theories and ambiguity theories. Under-specification theories (Anscombe 1964; Heinämäki 1974; Beaver & Condoravdi 2003; Krifka 2010), observing that there is an entailment relation between the two “readings” of (1a), posit that (1a) is not actually ambiguous. On this view, (1a)’s only LF has weak truth-conditions (“Emma left sometime after Hector started being happy”), truth-conditions which are verified both by after-start contexts like Scenario B and by after-finish contexts like Scenario C. In ambiguity theories (Condoravdi 2010; Rett 2020), by contrast, (1a) has an LF corresponding to the after-finish reading, and there is a separate mechanism that derives the after-start reading. As an exemplar, consider Rett’s (2020) ambiguity theory. Rett (2020) proposes that in the absence of coercion, *p before q* has a before-start meaning, and *p after q* has an after-finish meaning. To account for (1a)’s after-start reading, Rett assumes that a mechanism of **inchoative coercion** (Binnick 1991; Brennan & Pytkänen 2010) can apply to *Hector was happy* and coerce it into a change-of-state (*Hector started being happy*). In other words, *after Hector was happy* (for Rett) means “after Hector’s happiness ended” in the absence of coercion, but it can be coerced into meaning “after Hector’s happiness started.”

Among the theories of *before/after* that have considered the data with accomplishments, we observe a similar under-specification vs. ambiguity split for (2a). Under-specification theories of (2a) (Heinämäki 1974; Beaver & Condoravdi 2003; Condoravdi 2010), appealing to differences in the meanings of telic vs. atelic predicates, posit that (2a)’s only LF corresponds to the before-finish reading, with the stronger reading falling out as a subcase. By contrast, Rett’s (2020) ambiguity theory posits that the before-start reading surfaces by default; the before-finish reading only arises when we perform **completive coercion**, i.e. coerce *Jackson climbed the mountain* into an achievement (*Jackson finished climbing the mountain*).¹ Regarding theories that do not consider (2)–Anscombe 1964 and Krifka 2010—I show in section 2.1 that their under-specification approach to (1) can be easily extended to (2), putting them on a par with other under-specification theories.

To motivate her approach, Rett (2020) claims that inchoative and completive coercion are independently needed to deal with other data, such as (3). In (3a), an atelic predicate (*Hector was happy*) has a change-of-state meaning (*Hector started being happy*), which we can capture by assuming that the *within*-modifier forces inchoative coercion to apply to *Hector was happy* (Binnick 1991; Brennan & Pytkänen 2010). For (3b), Rett (2020) notes that it has a reading paraphraseable as “Jackson finished climbing the mountain at 7pm sharp,” and she accounts for

¹Rett (2020) claims that existing theories “cannot predict” (p. 8) the before-start reading of (2a), but this is, strictly speaking, not true: since the before-finish reading entails the before-start reading, any theory that can account for the before-finish reading also predicts the before-start reading. Nonetheless, Rett (2020) is indeed the first person to posit that (2a) is genuinely ambiguous, which is perhaps what she is alluding to in this passage.

this by assuming that the *at*-adverbial forces completive coercion to apply.

- (3) a. Within an hour, Hector was happy.
- b. Jackson climbed the mountain at 7pm sharp.

The debate between under-specification and ambiguity theories remains unsettled. On the empirical side, much of the debate has centered around NPI-licensing facts, where both types of theories have advantages and disadvantages (see discussion in Rett 2020). On the conceptual side, Rett’s (2020) appeals to (3a-b) could be construed as evidence for her position if coercion-based theories of these cases are indeed correct, but the necessity of appealing to coercion in (3a) has come into question (Krifka 2010).

The present study contributes to the debate by testing several online processing predictions that Rett’s (2020) theory makes and under-specification theories do not. Rett’s (2020) theory makes distinctive online processing predictions because coercion has a distinctive psycholinguistic profile (Todorova et al. 2000; Brennan & Pytkänen 2008; Brennan & Pytkänen 2010; Lukassek et al. 2017). In particular, under standard assumptions, parsers do not coerce by default and thus show processing slowdowns in a coercion sentence once they know that coercion is required to resolve a mismatch with an adverbial or with contextual knowledge. Using this perspective on coercion, previous studies have tested coercion vs. non-coercion theories of various data, arguing for coercion if the claimed coercion cases have an online cost (Piñango et al. 1999, 2006; Todorova et al. 2000; McElree et al. 2001; Traxler et al. 2002, 2005; Brennan & Pytkänen 2008, 2010; Lukassek et al. 2017) and arguing against coercion otherwise (Pickering et al. 2006; Lukassek et al. 2017). Indeed, Brennan & Pytkänen (2010) have previously found a processing cost for (3a), which can be seen as providing indirect support for Rett’s proposal. At the same time, no prior experimental work has directly examined the prospect of coercion in *before-/after*-clauses or the notion of completive coercion in general.

Once we take past findings on coercion into account, we see that if Rett’s theory is correct, parsers who see (1a) in an after-start context or (2a) in a before-finish context—that is, parsers who see (1a)/(2a) in contexts where the only available reading is one that involves coercion—should show processing slowdowns upon realizing that coercion is necessary. Additionally, given Rett’s claim that (3a-b) also involve coercion, she predicts that they should exhibit coercion-related slowdowns too.

We test these predictions via four self-paced reading experiments. In Experiments 1a-b, we investigate whether sentences like (3a-b) have the processing profile predicted by coercion-based theories, finding that while (3b) is costly in same way as other coercion cases, (3a) does not incur a processing cost at all. The latter result constitutes a replication failure vis-à-vis Brennan & Pytkänen (2010).

After Experiment 1, we follow up on the completive and inchoative coercion results in turn. Starting with completive coercion: since Experiment 1b finds an

online cost of completive coercion for (3b), Rett would expect that (2a)’s before-finish reading should be costly too. We test this prediction in Experiment 2 and find that the before-finish reading is indeed costly. Turning to inchoative coercion, Experiment 3 addresses one possibility for why Experiment 1a did not replicate Brennan & Pykkänen (2010), but we still find no cost for (3a). Finally, Experiment 4 tests whether (1a)’s after-start reading requires coercion, finding that after-start readings of *after* have the same sort of cost as before-finish readings of *before*.

Taken together, we argue that our results are compatible with Rett’s (2020) theory, though fully accounting for our results would require her to make further amendments (which we will discuss later). Though our results require further assumptions on the part of Rett (2020), we argue that they align better with Rett (2020) than with other existing theories, which do not predict asymmetries like those observed in Experiments 2 and 4.

The rest of this paper is structured as follows. In section 2, we provide further background on under-specification vs. ambiguity theories and on aspectual coercion. Section 3 lays out the predictions made by Rett’s (2020) theory in more detail. Sections 4-7 cover the methodology, results, and takeaways from Experiments 1-4, and Section 8 consists of a general discussion of the results and their consequences.

2 Background

2.1 *Before* and *after*: Under-specification or ambiguity?

In this section, we contrast an under-specification account with Rett’s (2020) theory. As an exemplar of an under-specification approach, we focus on Krifka (2010). We do so both because it is the most recent proposal of this type and because its appeal to implicature provides a useful point of comparison with Rett’s use of coercion. We defer discussion of Condoravdi (2010) until the General Discussion (section 8.3), since that proposal shares many features with Rett’s (2020) account but does not make explicit processing predictions. We do not discuss Anscombe (1964), Heinämäki (1974), or Beaver & Condoravdi (2003) separately, as these proposals have largely been superseded by later work.

2.1.1 An under-specification theory of *before/after* (Krifka 2010)

Under Krifka’s (2010) theory, the sentences in (1) receive the truth-conditions in (4), where n is the time of evaluation and $t' \leq t$ iff the interval t' precedes or overlaps the interval t . For ease of comparison, the contribution of the *before*-clause in (4a) is shown in boldface, as is the contribution of the *after*-clause in (4b). Note that Krifka (2010) seems to assume, as is standard, that stative predicates like *Hector was happy* are true at moments (i.e. time-points).

- (4) a. $\llbracket \text{Emma left before Hector was happy} \rrbracket^n = 1$ iff
 $\exists t [t < n \wedge \text{Emma left at } t \wedge$
 $\neg \exists t' [t' \leq t \wedge t' < n \wedge \text{Hector is happy at } t']]$
b. $\llbracket \text{Emma left after Hector was happy} \rrbracket^n = 1$ iff
 $\exists t [t < n \wedge \text{Emma left at } t \wedge$
 $\exists t' [t' \leq t \wedge t' < n \wedge \text{Hector is happy at } t']]$

These meanings capture the truth-conditional asymmetry with atelic predicates. To see this, consider the scenarios in Figure 1 again.² (4a) is only true in the before-start scenario (Scenario A), as desired: that is the only scenario where **none** of the runtime of Hector’s happiness precedes or overlaps the time at which Emma left. Turning to (4b), it correctly comes out true in Scenarios B and C: in both scenarios, **some** of the runtime of Hector’s happiness precedes or overlaps Emma’s departure.

While the meanings in (4) capture the basic asymmetry with atelic predicates, Krifka (2010):921 notes that (4b) does not quite reflect how *after* is typically understood. In particular, if Emma left at 6pm and Hector was happy 6pm-8pm, *Emma was happy after Hector was happy* is readily judged false, but Krifka’s theory predicts that it should be true. After all, we can instantiate both t and t' in (4a) with 6pm and get a true statement (since $6\text{pm} \leq 6\text{pm}$).

Krifka (2010) has a two-pronged response to this problem, a problem which stems from his use of $\leq$ rather than $<$ in (4b). First, Krifka (2010) notes that there are some cases in which *after* does clearly encode the $\leq$ relation: (5), for example, can be true even if the matrix and eventuality are simultaneous.

- (5) The light went off after she tripped the switch.

Second, to explain why *after* is usually not understood as encoding $\leq$, Krifka (2010) posits that *after* competes with *when*. For example, the idea is that in a context where Emma left at 6pm and Hector was happy 6-8pm, (4b) is literally true but has the stronger true alternative (6), making (4b) degraded in this context.

- (6) Emma left when Hector was happy.

Note that Krifka (2010) posits competition between all temporal connectives (*before*, *after*, *while*, *when*), not just between *after* and *when*. Hence, *before*- and *after*-sentences carry a variety of cancellable implicatures under Krifka’s (2010) theory; see section 3 for further discussion.

Krifka (2010) happens not to consider the judgments with accomplishments (2); however, his strategy for dealing with the atelic data can be extended to (2) with one small tweak, and once we do this, Krifka’s (2010) theory is empirically on a par with theories that have considered (2). To see this, let us start by looking at

²In what follows, I assume that the events depicted in Figures 1-2 take place prior to n .

the truth-conditions Krifka’s theory currently predicts for (2), shown in (7). Krifka (2010) does not state any assumptions about the meanings of accomplishments, but I will assume for concreteness that if Jackson climbed the mountain 9am-9pm, then “Jackson climbed the mountain at t ” is only true of the interval [9am,9pm] (not any subintervals). This is a standard claim about accomplishments (Dowty 1979), one that Krifka himself has adopted in his work on lexical aspect (Krifka 1998).

- (7) a. $\llbracket \text{Emma left before Jackson climbed the mountain} \rrbracket^n = 1$ iff
 $\exists t [t < n \wedge \text{Emma left at } t \wedge$
 $\neg \exists t' [t' \leq t \wedge t' < n \wedge \text{Jackson climbed the mountain at } t']]$
 b. $\llbracket \text{Emma left after Jackson climbed the mountain} \rrbracket^n = 1$ iff
 $\exists t [t < n \wedge \text{Emma left at } t \wedge$
 $\exists t' [t' \leq t \wedge t' < n \wedge \text{Jackson climbed the mountain at } t']]$

As it stands, the theory wrongly predicts that accomplishments should exhibit the same truth-conditional asymmetry as atelics. To see this, let’s again suppose that Jackson climbed the mountain 9am-9pm and consider the three scenarios in Figure 2 for when Emma left. Jackson’s climb precedes or overlaps Emma’s departure in Scenarios B and C, but not in Scenario A. Accordingly, (7a) comes out true in Scenario A only, while (7b) comes out true in Scenarios B and C. As established in section 1, these predictions are wrong for both sentences: the *after*-sentence cannot in fact be judged true in Scenario B, while the *before*-sentence can.

To remedy things, all a proponent of Krifka 2010 needs to do is to redefine “ $t' \leq t$ ” so that it means “all parts of t' precede or overlap t ” rather than “ t' precedes or overlaps t .” Now, (7a) comes out true not just in Scenario A but also in Scenario B, because not all parts of Jackson’s climb precede or overlap Emma’s departure in Scenario B. For the same reason, Scenario B now falsifies (7b), although (7b) is still true in Scenario C, as desired.³ This tweak to the definition of $\leq$ does not affect the theory’s good predictions about atelic predicates like *Hector was happy*: such predicates are true at time-points, and points do not have proper parts. Hence, the old and new definitions of $\leq$ give the same results for (4).

2.1.2 Rett (2020)

Having presented an example of an under-specification theory, we now contrast it with Rett 2020. Rett’s entries for *before* and *after* are given in (8). Note that Rett (2020), unlike Krifka (2010), treats propositions as denoting time-intervals directly; *Hector was happy*, e.g., denotes the total run-time of Hector’s happiness.

³Even after this tweak, (7b)–like (4b)–appears to make the wrong prediction in a particular edge case. In particular, if Emma left at the exact moment Jackson climbed the mountain, (7b) comes out true even though the sentence is readily judged false. Once again, Krifka (2010) could deal with this case by appealing to competition with *Emma left when Jackson climbed the mountain*.

- (8) a. $\llbracket A \text{ before } B \rrbracket = 1$ iff $\exists t \in A [t < \text{MAX}(B_{<})]$
 b. $\llbracket A \text{ after } B \rrbracket = 1$ iff $\exists t \in A [t > \text{MAX}(B_{>})]$

Under Rett’s (2020) theory, both *before* (8a) and *after* (8b) invoke a function MAX in their lexical entries, the definition of which is given in (9)—note that R is a variable that can take on the values $<$ or $>$. Informally, (9) makes it so that $\text{MAX}(B_{<})$ is the first time-point in the interval B , while $\text{MAX}(B_{>})$ is the last time-point in B . Rett (2020) calls this function MAX because it can be thought of as picking out the maximally informative member of the interval X with respect to a given ordering.⁴ For example, in *A before B*, the first time-point within B (i.e. $\text{MAX}_{<}(B)$) is maximally informative in the sense that if A is before the first time-point in B , A must be before every other time-point in B .

$$(9) \quad \text{MAX}(X_R) = \iota x [x \in X \wedge \forall x' \in X [x \neq x' \rightarrow x R x']]$$

By themselves, the entries in (8) predict most but not all of the possible readings. To see this, let us start by considering the our running atelic example, using our context where Hector was happy 6pm-8pm.

- (10) a. $\llbracket \text{Emma left before Hector was happy} \rrbracket$
 $= 1$ iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t < \text{MAX}([6\text{pm}, 8\text{pm}]_{<})]$
 $= 1$ iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t < 6\text{pm}]$
 b. $\llbracket \text{Emma left after Hector was happy} \rrbracket$
 $= 1$ iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t > \text{MAX}([6\text{pm}, 8\text{pm}]_{>})]$
 $= 1$ iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t > 8\text{pm}]$

We see that (10a) comes out true iff Emma left sometime before Hector started being happy, while (10b) comes out true iff Emma left sometime after Hector stopped being happy. These meanings correspond to intuitions in the case of the *before*-sentence, but the truth-conditions for the *after*-sentence are too strong. In particular, (10b) captures the after-finish reading of the sentence but not the after-start reading: (10b) is false in an after-start scenario like Figure 1’s Scenario B.

To capture the full range of readings, Rett (2020) posits that whenever we judge (10b) true in an after-start scenario like Scenario B, we are coercing *Hector was happy* into having a change-of-state meaning (*Hector started being happy*). Rett (2020) refers to this mechanism as **inchoative coercion**, following standard terminology (she specifically cites Dölling 2014). Formally, Rett (2020) posits that while *Hector was happy* denotes the full runtime of Hector’s happiness in the absence of coercion, it denotes the start-point of Hector’s happiness when inchoative coercion applies. This mechanism, Rett assumes, only applies to atelic predicates.

⁴The notion of maximal informativity is well-established; Rett (2020) is especially inspired by work in degree semantics involving this notion (e.g. Rullmann 1995; Beck & Rullmann 1999).

(11a) shows the meaning of *Emma left after inchoative(Hector was happy)* in our example where he was happy 6pm-8pm. While the version of the sentence without coercion is only true in an after-finish scenario, (11a) is also true in an after-start scenario. Note that nothing in the derivation would go wrong if inchoative coercion were to apply within an atelic *before*-clause (11b), but it would be vacuous: (11b) is equivalent to the version of the sentence without coercion.

- (11) a. $\llbracket \text{Emma left after inchoative(Hector was happy)} \rrbracket$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma was happy} \rrbracket [t > \text{MAX}(\text{inchoative}([6\text{pm}, 8\text{pm}])_{>})]$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t > \text{MAX}([6\text{pm}]_{>})]$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t > 6\text{pm}]$
 b. $\llbracket \text{Emma left before inchoative(Hector was happy)} \rrbracket =$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t < \text{MAX}([6\text{pm}]_{<})]$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t < 6\text{pm}]$

Turning to our telic example, (12) shows the predicted truth-conditions in our set-up where Jackson climbed the mountain 9am-9pm.

- (12) a. $\llbracket \text{Emma left before Jackson climbed the mountain} \rrbracket$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t < \text{MAX}([9\text{am}, 9\text{pm}]_{<})]$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t < 9\text{am}]$
 b. $\llbracket \text{Emma left after Jackson climbed the mountain} \rrbracket$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t > \text{MAX}([9\text{am}, 9\text{pm}]_{>})]$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t > 9\text{pm}]$

(12b) is only true in an after-finish scenario, as desired, but (12a) is too strong. In particular, (12a) captures the before-start reading of the sentence but not the before-finish reading. To capture the full range of readings, Rett (2020) posits that whenever we judge (12a) true in a before-finish scenario, we are coercing *Jackson climbed the mountain* into its telos. Rett (2020) calls this process **completive coercion** and assumes that it can only apply to telic predicates.⁵

(13a) shows the meaning of *Emma left before completive(Jackson climbed the mountain)* in our example where Jackson climbed the mountain from 9am-9pm. While the version of the sentence without completive coercion is only true in Scenario A, (13a) is also true in Scenario B. Completive coercion can in principle apply within a telic *after*-clause as well (13b), but it would be vacuous: (13b) is equivalent to the version of the sentence without completive coercion.

- (13) a. $\llbracket \text{Emma left before completive(Jackson climbed the mountain)} \rrbracket$
 $= 1 \text{ iff } \exists t \in \llbracket \text{Emma left} \rrbracket [t < \text{MAX}(\text{completive}([9\text{am}, 9\text{pm}])_{<})]$

⁵Rett (2020) cites Dölling (2014) for the notion of completive coercion, but as far as we can see, Rett's (2020) notion of completive coercion is different from Dölling's.

- = 1 iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t < \text{MAX}([9\text{pm}]_<)]$
- = 1 iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t < 9\text{pm}]$
- b. $\llbracket \text{Emma left after } \mathbf{completive}(\text{Jackson climbed the mountain}) \rrbracket =$
- = 1 iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t > \text{MAX}([9\text{pm}]_>)]$
- = 1 iff $\exists t \in \llbracket \text{Emma left} \rrbracket [t > 9\text{pm}]$

Rett (2020) provides several motivations for her approach. First, she observes (based on a survey of 17 languages) that all languages seem to allow before-start readings of *before* and after-finish readings of *after*, but not all languages allow before-finish readings of *before* or after-start readings of *after*. This fact arguably favors a theory where before-start readings of *before* and after-finish readings of *after* are “basic” and other readings are derived via a language-specific mechanism. Second, Rett (2020) notes that Tagalog distinguishes before-start and before-finish readings using overt morphology in a way that mirrors completive coercion. Rett (2020) also finds that many languages use an inchoative marker in atelic *after*-clauses, but she does not find one where the inchoative marker changes the meaning.

Third, Rett (2020) emphasizes that inchoative and completive coercion may be required independently. Inchoative coercion has been invoked for a variety of other data (Binnick 1991; Brennan & Pylkkänen 2010; Dölling 2014), including sentences where an atelic predicate occurs alongside a *within*-modifier (14a). These sentences have a change-of-state meaning (*within an hour Hector became happy*), which is expected if we assume that inchoative coercion applies in (14a). To motivate completive coercion, Rett (2020) notes that (14b) can have a “completive” reading (*Jackson finished climbing the mountain at 7pm*) and accounts for this by assuming that completive coercion applies to the accomplishment in (14b).

- (14) a. Within an hour Hector was happy.
- b. Jackson climbed the mountain at 7pm sharp.

Taken together, these arguments suggest that an ambiguity theory may have conceptual advantages (though see Krifka 2010 for the conceptual advantages of under-specification theories). However, there is a dearth of English-internal empirical arguments for the claim that *before*-/*after*-clauses, on certain readings, have coercion mechanisms analogous to aspectual markers in other languages; hence, the debate between under-specification and ambiguity theories is not fully settled.

2.2 Past literature on coercion

Having summarized the debate on how the truth-conditions of *before*-/*after*-sentences are derived, we now turn to the other main body of work this paper engages with: the literature on coercion.

There are a variety of data for which coercion-based theories have been proposed (see Dölling 2014 for an overview). In addition to the comparatively little-studied cases discussed so far, two very well-studied cases involve sentences like (15a) and (15b). In (15a), an entity-denoting expression (*the book*) takes on an eventive meaning (‘reading the book’) in the presence of a verb like *finish*; in (15b), a punctual predicate (*jump*) takes on a repetitive meaning (*jump repeatedly*) in the presence of a *for*-adverbial.

- (15) a. I finished the book that you told me to read.
 b. For an hour, the zebra jumped in its enclosure.

Coercion is generally conceptualized as a repair mechanism (following Moens & Steedman 1988, a.o.). In some cases, the issue that coercion is performed to repair is a semantic mismatch. For example, proponents of a coercion theory of (15a) assume that there is a type mismatch between certain verbs (*begin*, *finish*, etc.) and entity-denoting expressions like *the book*; to resolve the mismatch, we insert a **complement coercion** operator, which coerces *the book* into having an eventive meaning (Jackendoff 1997, McElree et al. 2001, Traxler et al. 2002, Traxler et al. 2005, and much other work, *pace* Piñango & Deo 2016). In other cases, the issue that coercion is performed to repair can either be thought of as a semantic mismatch or as a pragmatic violation. For example, some proponents of a coercion theory of (15b) posit that there is a semantic mismatch between the durative adverbial *for an hour* and the punctual predicate *jump* (de Swart 1998). Others suggest that the semantic derivation for (15b) is mismatch-free, but the resulting meaning (“the zebra jumped once, and it lasted for an hour”) is infelicitous because it conflicts with world knowledge (Dölling 2014). Both variants of the theory propose that (15b) is repaired by a process of **iterative coercion**, which forces (15b) to be reinterpreted as *For an hour, the zebra jumped repeatedly...*

There are yet other claimed coercion cases where the issue that motivates repair only be thought of as pragmatic in nature. Consider (16), from Dölling (2014).

- (16) Mike swam for twenty years.

Dölling (2014) observes that in most contexts, the atelic predicate *swam* in (16) will receive the habitual meaning “was a swimmer” rather than a literal one. It would be implausible to posit a semantic mismatch between *swim* and *for*-adverbials (which are prototypical modifiers of atelic predicates). Rather, Dölling (2014) suggests, the habitual reading arises because the literal meaning, “Mike swam non-stop for twenty years straight,” conflicts with world knowledge, necessitating coercion of *Mike swam* into a habitual meaning.

Coercion theories make real-time processing predictions (Todorova et al. 2000; Brennan & Pylkkänen 2008; Lukassek et al. 2017). In particular, if a sentence is

underlyingly ill-formed or infelicitous and can only be fixed with a re-interpretation mechanism (i.e. coercion), we expect that parsers encountering such sentences will show processing slowdowns upon realizing that coercion is necessary; these slowdowns reflect the cognitive operations involved in re-interpreting the sentence. This prediction of coercion accounts holds regardless of whether the problem that necessitates coercion is semantic or pragmatic (Brennan & Pylkkänen 2008).

Because coercion theories make processing predictions, psycholinguistic studies have been an important testbed for coercion theories of various phenomena. Most such studies look at claimed cases of complement coercion or iterative coercion. The theory of complement coercion predicts that sentences like (15a) should be harder to process than controls (e.g. *I read the book...*) at or just after *book*, as parsers have seen *finish* + an entity-denoting complement and thus know that complement coercion is required. All studies on (15a) known to us (starting with McElree et al. 2001) bear out this prediction, though Delogu et al. (2017) have recently argued that most (but not all) of the processing effect stems from the fact that *book* has higher surprisal (i.e. is less predictable) in *finish the book* vis-à-vis *read the book*.⁶

Similarly, the theory of iterative coercion predicts that sentences like (15b) should be harder to process than controls at or one word after *jumped*, as parsers have seen *for an hour* + a punctual predicate and thus know that coercion is required. Most studies bear out this prediction and thus conclude in favor of the coercion approach (Piñango et al. 1999, 2006; Todorova et al. 2000; Brennan & Pylkkänen 2008). Conversely, Pickering et al. (2006) do not find a cost and accordingly argue for an alternative, under-specification-based view.

While iterative and complement coercion have been the focus of most psycholinguistic coercion studies, the processing of other types of coercion has been studied too. Most crucially for our purposes, Brennan & Pylkkänen (2010) have compared the processing of claimed inchoative coercion cases like (17a) to controls like (17b).

- (17) a. Within 1 week the dragon detested the town’s people.
- b. Obviously the dragon detested the town’s people.

Brennan & Pylkkänen (2010) find a processing slowdown for (17a) one word after the main verb, as predicted by the inchoative coercion theory.

3 Deriving processing predictions from formal theories

Having summarized existing literature on coercion and on *before/after*, we now spell out the processing predictions of Rett’s (2020) theory and contrast them with those of Krifka (2010).

⁶We thank an anonymous reviewer for informing us about Delogu et al. 2017.

3.1 Rett’s (2020) predictions

To spell out Rett’s (2020) predictions, we look at each of the four kinds of coercion cases she posits in turn. Following past literature (e.g. Binnick 1991, Brennan & Pytkänen 2010), we can spell out the coercion account of (18) in terms of a semantic mismatch. In particular, the claim is that *within*-modifiers select for telic predicates and thus clash compositionally with atelic predicates. Inchoative coercion repairs the mismatch because **inchoative**([Jessica was mad in office hours]), which has the change-of-state meaning “Jessica became mad in office hours,” is telic.

- (18) Within an hour, Jessica was mad in office hours.

This account predicts that (18) should be hard to process at or one word after *was*, as parsers have seen a *within*-modifier + the beginning of a stative and thus know that inchoative coercion is required. **Experiment 1a** tests this prediction, and **Experiment 3** tests a similar prediction for subject-experiencer verbs (*within an hour, Ben loved Sue*), the predicates used by Brennan & Pytkänen (2010).

One can spell out Rett’s (2020) completive coercion account of sentences like (19) in terms of a semantic mismatch or a pragmatic violation. If one opts for the semantic route, one would say that there is a semantic mismatch between the durative VP *climbed the mountain* and the punctual adverbial *at 7pm sharp*. If one opts for the pragmatic route, one would say that the semantic derivation for (19) is mismatch-free but that the resulting meaning (“at 7pm sharp, Jackson both started and finished his climb”) conflicts with world knowledge. In either case, completive coercion resolves the mismatch because **completive**([Jackson climbed the mountain]), which means “Jackson finished climbing the mountain,” is punctual.

- (19) At 7pm sharp, Jackson climbed the mountain.

Both the semantic and pragmatic variants of the theory predict that in a context where Jackson climbed one mountain and the climb ended at 7pm, (19) should incur a cost at or one word after *climbed*; at that point, the parser has seen *at 7pm*, has seen the start of an accomplishment, and knows the climb ended at 7pm, so they know that completive coercion is needed.⁷ **Experiment 1b** tests this prediction.

We cannot describe Rett’s (2020) theory of (20) in terms of a semantic mismatch, as the derivation for (20) does not crash without completive coercion, according to her: the coercion-less derivation merely has a different reading than the derivation with completive coercion. However, it is possible to frame her theory of (20) in pragmatic terms with the help of the assumption that completive coercion is only licensed when the before-start reading would conflict with contextual information.

⁷Note that atelic continuations (e.g. *climbed mountains*) are implausible since Jackson only climbed one mountain in the context.

In other words, when coercion occurs in (20), it is performed to repair a mismatch between the “standard,” before-start truth-conditions of (20) and the context.

(20) Emma left before Jackson climbed the mountain.

Under this theory, we would expect that in a context that is only compatible with the before-finish reading (and in which Jackson climbed one mountain), (20) should be hard to process at or one word after *climbed*. At that point, parsers have seen the start of an accomplishment in a *before*-clause but know that only a before-finish reading is compatible with the context; they thus spot the mismatch between (20)’s default truth-conditions and the context and know to coerce. Crucially, reading times at or one word after *climbed* in (20) should be longer in a before-finish context than in a before-start context, since coercion in (20) is not necessary in the latter context. **Experiment 2** tests this prediction.

We can reframe Rett’s (2020) theory of atelic *after*-sentences like (21) in a similar fashion. In particular, we can assume that the after-finish reading is the only reading available in a neutral context, and inchoative coercion is only licensed when the after-finish reading would conflict with contextual information.

(21) Dave was regretful after Lara feared the large dog.

Under this theory, we would expect that in a context that is only compatible with the after-start reading, (20) should be hard to process at or one word after *feared*; at that point, parsers will project a stative but know that only an after-start reading is possible, so they know to coerce. **Experiment 4** tests this prediction.

Two further comments on these predictions bear mention. First, a caveat: past research has found online costs for coercions that repair a semantic violation (e.g. complement coercion) and for coercions that repair a violation that can either be thought of as semantic or pragmatic (e.g. iterative coercion). No psycholinguistic research known to us looks at potential cases of coercion that must be thought of as resolving a pragmatic mismatch (e.g. (20)-(21) or the habitual reading of *Mike swam for twenty years*, see above), though Brennan & Pytkänen (2008) provide neural imaging evidence suggesting that the underlying deviance of iterative coercion sentences is pragmatic. As such, it remains unknown whether semantic and pragmatic coercion have the same psycholinguistic profile. However, past literature has generally assumed that they should incur a similar cost in behavioral paradigms like self-paced reading (Brennan & Pytkänen 2008), so the *prima facie* predictions of Rett’s (2020) theory are best stated as above.

Second, one might worry that it runs counter to the intent of Rett’s (2020) theory to assume that before-finish readings of *before* and after-start readings of *after* are only licensed when the other reading would conflict with the context. However, there are several indications in Rett’s work that coercion in *before*-/ *after*-clauses

(on her theory) is subject to a pragmatic condition of this sort. For one thing, Rett (2020):216 relates her theory of *before/after* to her theory of the “polar asymmetry” in comparatives (Rett 2015), a theory that relies on this sort of contextual licensing. The polar asymmetry for comparatives, observed by Seuren (1973), is shown in (22). While (22a) can only mean that Lucinda is over the speed limit, (22b) is ambiguous: it can mean that Lucinda is not over the speed limit (the <max reading), or that Lucinda is slower than the minimum speed allowed (the <min reading).

- (22) a. Lucinda is driving faster than allowed on this highway. ($\checkmark$ >max, $\times$ >min)
b. Lucinda is driving slower than allowed on this highway. ($\checkmark$ <max, $\checkmark$ <min)

Rett’s (2015) theory of (22b) rests on the claim that the stronger, <min reading is the only one available in a context compatible with both readings; the <max reading only arises in a context that is incompatible with the stronger reading.⁸ The fact that Rett (2020) highlights the parallels between (22) and the data with *before/after* suggests that the weak (i.e. coerced) readings of *before* and *after* only arise when the stronger reading is blocked (for her), like the weak reading of (22).⁹

Furthermore, Rett (2026) characterizes Rett 2020 in ways that imply a pragmatic licensing condition of this sort. For example, she at one point contrasts the coerced reading of atelic *after*-clauses with “the standard, non-coerced reading,” (p. 20), implying that the non-coerced reading is what arises by default. Similarly, she states that some contexts “instigate completive coercion” (p. 19) and others do not, with the non-coerced reading arising in contexts that do not instigate coercion.

In sum, we think that our interpretation of Rett (2020) is viable. There may be others, but what will be relevant going forward is that there is a plausible interpretation of Rett 2020 that predicts a processing asymmetry in all of our experiments.

3.2 Krifka’s (2010) predictions

As discussed in section 2.1.1, the two readings of atelic *after*-clauses are derived via the same LF under Krifka’s (2010) theory, and this strategy can easily be extended to telic *before*-clauses as well. Given this, it may seem that Krifka, in contrast to Rett, does not predict that the processing of (20)-(21) should be affected by context (as long as the sentence is true); after all, the LF is the same on both “readings.”

⁸The specific kind of <min-blocking context Rett (2015) focuses on is a context where there is no minimum speed. But the pragmatic principle she posits—“the property is interpreted with respect to the most informative bound available” (p. 20)—is a general one, and can naturally be interpreted as extending to contexts where the <min reading is blocked for other reasons.

⁹Rett (2015) actually does discuss touch on *before* and *after*, but she does not claim that *before* and *after* are ever ambiguous. Hence, the parallels between *before/after* and (22) actually shine through more strongly in Rett 2020 than in Rett 2015.

However, while the two readings of (20) and (21) have the same literal semantics (for Krifka), the two readings differ in the implicatures they violate (or lack thereof); this fact makes his processing predictions about (20) and (21) less clear. The implicatures in question stem from the competition Krifka (2010) posits between *before/after* and *while*. For example: while (20) is literally true in a before-finish scenario (e.g. Figure 2ii), using (20) in this context would, under Krifka’s theory, violate the implicature “it is not the case that Emma left while Jackson climbed the mountain;” by contrast, (20) violates no implicatures in a before-start context. Similar considerations apply to (21): using (21) in a after-start context would violate the implicature “it is not the case that Emma left while Hector was happy,” but using (21) in a after-finish context would respect this implicature.

The processing of implicature is a complicated matter: see, e.g., Huang & Snedeker (2009) for evidence that implicatures are costly, Grodner et al. (2010) for evidence of the opposite, and Degen & Tanenhaus (2015) for an intermediate view. Nonetheless, the fact that the two readings of (20) and (21) differ in their implicature profile (under Krifka’s theory) means that a processing asymmetry between these two readings would not be entirely unexpected for him.

Given this state of affairs, do Krifka and Rett make different processing predictions after all? As it turns out, while it is hard to say whether the two theories differ in their predictions for (20) and (21), their predictions clearly differ for examples with an atelic predicate in the matrix clause. For example, suppose Jackson climbed the mountain 9am-9pm and consider (23) in two scenarios: in Scenario A, Emma was nervous 6am-9pm, and in Scenario B, she was nervous 3pm-9pm.

(23) Emma was nervous before Jackson climbed the mountain.

Under Rett’s (2020) theory, coercion is not needed to make (23) true in Scenario A, since there is a part of Emma’s nervousness that precedes the start of Jackson’s climb. By contrast, the only way to make (23) true in Scenario B is to perform completive coercion. Hence, as discussed above, there is a viable interpretation of Rett’s theory that predicts that (23) should be harder to process in Scenario B.

Krifka’s (2010) theory, by contrast, does not draw a distinction between the two contexts. In both contexts, the literal meaning of the sentence is verified; furthermore, since both contexts involve temporal overlap, they both falsify the putative implicature $\neg$ *Emma was nervous while Jackson climbed the mountain*.¹⁰

Similarly, the two theories make different predictions about atelic *after*-clauses when the matrix predicate is also atelic. For example, suppose that Hector was happy 6pm-8pm and consider (24) in two scenarios: in Scenario A, Emma was happy 7pm-9pm, and in Scenario B, Emma was happy 7-8pm.

¹⁰These two contexts are also the same regarding the other implicatures that Krifka posits: both contexts respect the implicature $\neg$ *Emma was nervous after Jackson climbed the mountain*, and both contexts violate the implicature $\neg$ *Emma was nervous when Jackson climbed the mountain*.

(24) Emma was happy after Hector was happy.

In Scenario A but not in Scenario B, there is a part of Emma’s happiness that follows the end of Hector’s happiness; hence, Rett (2020) predicts that inchoative coercion is needed in Scenario B only. Conversely, Krifka (2010) does not distinguish between the two contexts: (24) is true in both, and since the contexts involve overlap to the same extent, they behave the same regarding the “not *while*” implicature. For these reasons, we will use sentences like (23)-(24) in our Experiments 2 and 4.

4 Experiment 1

We started testing the predictions of Rett’s (2020) theory by trying to replicate Brennan & Pytkänen (2010) and, in parallel, examining whether sentences with *at*-modifiers + accomplishments have the processing profile predicted by the completive coercion account. Collectively, these manipulations test Rett’s claim that inchoative and completive coercion are independently needed outside of *before/after*-clauses.

All of our experiments were pre-registered (Experiment 1: [click here](#), Experiment 2: [click here](#), Experiment 3: [click here](#), Experiment 4: [click here](#)). Data, materials, and analysis scripts can be found [here](#).

4.1 Methodology

Experiment 1 had two between-subjects sub-experiments, Experiments 1a and 1b, which focused on inchoative and completive coercion, respectively.

4.1.1 Participants

We recruited 220 self-reported native English speakers from Prolific (110 per sub-experiment), intending to use the subset of participants who met our inclusion criteria (see sec. 4.2) as our analyzed sample. All participants received £2.10.

For self-paced reading analyses, we excluded data from those who gave a rating of 3 out of 5 or higher on at least three out of our six ungrammatical/false fillers. 27 participants in Experiment 1a and 31 participants in Experiment 1b were excluded under this criterion.¹¹ We also excluded RTs that were >3 standard deviations from the mean for a given word type (following Brennan & Pytkänen 2010).¹²

¹¹This exclusion rate is quite high, especially since this criterion does not seem very strict. Despite the high exclusion rate, we will not arbitrarily tweak our exclusion criterion *post hoc*. One possible reason for the high exclusion rate is that Likert ratings are just not an ideal way to gauge attention, as even attentive participants might not all use a Likert scale in the same way. In fact, Brennan & Pytkänen (2010) (whose exclusion rate was lower) used binary judgments rather than Likert ratings. Hence, we tried switching to binary judgments for Experiment 2’s attention checks.

¹²We pre-registered that we would exclude RTs <150ms and >2500ms but ultimately decided

4.1.2 Materials

Experiment 1a. Experiment 1a was a conceptual replication of Brennan & Pylkkänen’s (2010) Experiment 1. Participants saw 26 trials, each of which proceeded in three phases. First, participants read a short context sentence-by-sentence. Then, they read a target sentence word-by-word using a self-paced reading interface. Finally, participants rated the naturalness of the target sentence given the context on a 5-point Likert scale.¹³ Following Brennan & Pylkkänen (2010), naturalness ratings were included because they allowed us to construct attention checks (see sec. 4.2), and because they can provide further insight into participants’ behavior.

Experiment 1a had four critical trial types (see **Table 1** for a sample item set). On inchoative-coercion trials (e.g. (a) in Table 1), participants read a context like (25a), followed by a target sentence with an atelic predicate + *within*-adverbial. On inchoative-control trials (e.g. (b) in Table 1), participants read a context like (25b), followed by a target sentence with an atelic predicate + *for*-adverbial. Trials like (b) are suitable non-coercive controls for (a) because *for*-adverbials are prototypical modifiers of atelic predicates, and as such no theory predicts that (b) should be difficult. Note that inchoative-coercion and inchoative-control trials used different contexts because we felt that, e.g., trying to construct a single context that verifies the target sentences in both (a) and (b) would result in something too convoluted.

In addition to trials like (a) and (b), we also included trials with complement-coercion target sentences (e.g. (c) in Table 1) and matched controls with non-coercive verbs (d). These two trial types were included as a sanity check: replicating past findings that (c)’s target sentence is costly (see section 2.2) helps show that our paradigm is sensitive enough to capture coercion effects. If we replicate past findings on complement coercion but find no comparable effect for another coercion type in a given experiment, the null result on the second coercion type cannot be chalked up to an overall inability to capture coercion effects within our paradigm.

- (25) a. **Context A:** One semester, Jessica took a class taught by a professor, Alex, whose lectures were extremely difficult to understand. Jessica had heard that Alex was more coherent in office hours, so she scheduled a 30 minute appointment with him. Alex did nothing but read one of his own articles to Jessica in full; by the fifteen-minute mark, Jessica voiced her annoyance about the professor’s behavior, but he paid her no mind and kept reading.
- b. **Context B:** Same first two sentences as Context A, then: Alex did nothing but read one of his own articles to Jessica in full during the first fifteen minutes, which annoyed her the whole time; she tried to

that it would be less arbitrary to use this Brennan & Pylkkänen-style criterion.

¹³Each trial unfolded next to a scene-setting picture (see Supplementary Materials for details).

get Alex to stop, but he paid her no mind and kept reading until the fifteen-minute mark.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Within fifteen minutes Jessica was mad in office hours.
	(b) Control	B	For fifteen minutes Jessica was mad in office hours.
Complement	(c) Coercion	A/B	Alex continued his article despite Jessica’s protest.
	(d) Control	A/B	Alex read his article despite Jessica’s protest.

Table 1: Sample stimuli for Experiment 1a.

Three differences between Experiment 1a’s design and Brennan & Pykkänen (2010) bear mention. First, we departed from Brennan & Pykkänen (2010) in our use of contexts. This was because most of our experiments needed contextual support to single out the relevant reading, and we wanted to minimize differences across experiments. Second, Brennan & Pykkänen (2010) used a variety of adverbials in their control trials, whereas we only used *for*-adverbials. We made this choice because the *within*-adverbial vs. *for*-adverbial contrast is as minimal as possible and because Brennan & Pykkänen (2010) report some noisiness that may have resulted from the different adverbials. Third, Brennan & Pykkänen’s inchoative coercion trials used subject experiencer verbs, not *be*-statives. The theory of inchoative coercion does not expect a processing difference between these two predicates; Brennan & Pykkänen (2010) used subject experiencers because they had an additional subject- vs. object-experiencer manipulation. We used *be*-statives because it was easier to create contexts for these predicates.

In addition to the four critical trial types, Experiment 1a had two types of fillers: ungrammatical fillers, which used target sentences with violations of selectional restrictions (e.g. *Karen meowed the phone close to midnight*), and grammatical fillers, which used a mixture of true and false target sentences.

Experiment 1b. The paradigm and structure of Experiment 1b were the same as Experiment 1a, but the target sentences and contexts differed. Experiment 1b had four critical trial types (see **Table 2**) and two types of contexts: those with an *at*-adverbial in the final sentence (e.g. ending A in (26)) and those with a phrase like *X minutes after...* in the final sentence (ending B). On completive-coercion trials (e.g. (a) in Table 2), participants read a context whose final sentence had an *at*-adverbial, followed by a target sentence with an accomplishment + *at*-adverbial. On completive-control trials (e.g. (b) in Table 2), participants read a context whose final sentence had *X minutes after...*, followed by a target sentence with an accomplishment + *in*-adverbial; *in*-adverbials are prototypical modifiers of accomplishments. Beyond these trials, we also included complement-coercion trials (c) and matched controls (d) as a sanity check. The contexts for these trials used

a combination of endings A and B, e.g. *...at 9pm, five minutes after Hector made the bet.*¹⁴

- (26) Context: Hector is a lifelong hiker and his daughter Emma wants to learn more about the outdoors, so they went on a hiking trip. On the first night, Hector bragged about how quickly he could set up a tent, and Emma bet him that he couldn't do it in ten minutes. Hector started building the tent at 8:55pm, and the tent was fully set up (**A**) at 9pm. (**B**) five minutes after Emma made the bet.

Trial type		Context	Target sentence
Compleitive	(a) Coercion	A	At 9pm sharp Hector built the humble tent.
	(b) Control	B	In five minutes Hector built the humble tent.
Complement	(c) Coercion	A + B	Hector finished the tent in front of Emma.
	(d) Control	A + B	Hector built the tent in front of Emma.

Table 2: Sample stimuli for Experiment 1b.

The contexts here ensured that participants interpreted (a) with compleitive coercion (*at 9pm, Hector finished*) rather than in some other way, e.g. with a world knowledge violation (*at 9pm, Hector both started and finished*).

We implemented the task in PCIBex (Schwarz & Zehr 2021). Sample trials from Experiments 1a and 1b are available [here](#) and [here](#), respectively.

4.1.3 Procedure

After providing consent, participants saw an instruction text and a practice trial (accessible [here](#)), which was the same in Experiments 1a and 1b. Afterwards, participants in Experiment 1a were sorted into one of eight lists, each of which had 16 critical trials and 10 fillers (4 true, 2 false, 4 ungrammatical). The 10 fillers were the same across lists. For the critical trials, we created 16 item sets like the one in Table 1 (each using a different *be*-stative) and used a Latin square design where each participant saw one trial from each item set. For example, participants in lists A and B saw the inchoative-coercion trial from Table 1, participants in lists C-D saw the inchoative-control trial from this item set (b), participants in list E-F saw the complement-coercion trial (c), and participants in lists G-H saw the complement-control trial (d). Due to this Latin square design, each participant saw 4 examples of each of the 4 critical trial types. The lists in Experiment 1a also

¹⁴While we could have used a combination of endings A and B on all trial types, this might have made the first few words of (b)'s target sentence easier to process than (a)'s (because *five minutes* was read more recently than *at 9pm*), and we wanted to avoid pre-verbal differences.

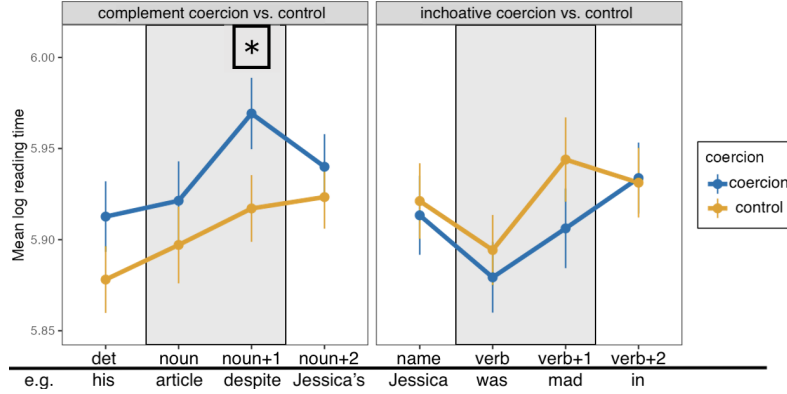


Figure 3: Mean logRTs in Experiment 1a. The regions of interest are shown in grey. Error bars are ± 1 standard error of the mean (SEM). Asterisks (or lack thereof) reflect the results of our statistical models.

differed as to whether complement-coercion trials (and controls) used contexts like Context A (lists A, C, E, and G) or ones like Context B (lists B, D, F, and H).

Participants in Experiment 1b were sorted into one of four lists, each of which had the same trial breakdown as Experiment 1a. As in Experiment 1a, the 10 fillers were the same across lists; for the critical trials, we created 16 items sets like the one in Table 2 (each centered around a different accomplishment) and used a Latin square design. Note that Experiment 1b contains fewer lists overall because it lacked a context manipulation on complement-coercion trials and controls. In all of our experiments, trials were presented in fully random order.

4.2 Results

4.2.1 Experiment 1a

Figure 3 shows RT results for the analyzed subset of Experiment 1a’s data; note that we opted to use log-transformed RTs rather than raw RTs in all of our experiments, as the former were more normal. Figure 3’s left panel shows logRTs for complement coercion trials like (c) in Table 1 and matched controls like (d) in and around the region of interest for complement coercion. Recall that following past literature, we expect longer RTs for complement coercion trials relative to controls at the direct object noun (the critical word) or noun+1. Figure 3’s right panel shows logRTs for inchoative coercion trials (e.g. (a) in Table 1) and matched controls (b) around the region of interest for inchoative coercion; the inchoative-coercion theory expects longer RTs for inchoative coercion trials vis-à-vis controls at verb or verb+1.

We tested our RT hypotheses using a pair of mixed-effects linear regression models with a structure similar to the models used in Brennan & Pytkänen’s

(2010) Experiment 1. We fit our models using the *lmerTest* package (Kuznetsova et al. 2017) in R (R Core Team 2025), an extension of the *lme4* package (Bates et al. 2015). Both models used data from all four critical trial types, with the first model being fit on critical word logRTs (verb in trials like (a-b) in Table 1; direct object noun in trials like (c-d)), and the other model being fit on critical+1 RTs (verb+1 in (a-b); noun+1 in (c-d)). The models predicted logRT from trial type and three covariates: word length (in characters), log lexical frequency according to the English Lexicon Project (Balota et al. 2007), and previous word logRT. These covariates were also used in Brennan & Pykkänen’s models.¹⁵

A reviewer notes that surprisal would have also been good to include, especially given past findings about surprisal and the complement coercion effect (Delogu et al. 2017). See **Appendix** for a version of our analyses that includes this (non-pre-registered) covariate.

In our models, we started with the maximal random-effects structure justified by the design (by-participant and by-item random intercepts and slopes for trial type), reducing stepwise until we got a converging, non-singular model.¹⁶ The final critical-word model had random intercepts for participants and items; the critical+1 model had by-participant random slopes too.¹⁷ Since we probed for the same effect(s) in two regions, we used a Bonferroni-corrected α threshold of 0.025 (i.e. $0.05/2$).

Using the *fauX* package (DeBruine 2025), we coded the “trial type” variable so as to extract three contrasts: (I) inchoative-coercion vs. inchoative-control; (II) complement-coercion vs. complement-control; (III) inchoative-coercion vs. complement-coercion. We are only interested in contrasts (I) and (II), but we had to include a third contrast in the model because the “trial type” variable had four levels. We only interpreted the p-values corresponding to contrasts (I) and (II).

In the model on critical word logRTs, contrasts (I)-(II) were not significant. The model on critical+1 logRTs found that complement-coercion trials had significantly longer logRTs than matched controls at noun+1 ($\beta = 0.038$, $SE = 0.015$, $p = 0.01$), though contrast (II) was still insignificant ($\beta = -0.023$, $SE = 0.01$, $p = 0.12$).

Figure 4 shows mean ratings for the four critical trial types (note that ungrammatical fillers were rated 1.72 on average). To analyze these data, we fit a pair of mixed-effects ordinal logistic regression models using the *clmm()* function from the

¹⁵The models in Brennan & Pykkänen’s (2010) Experiment 1 also had length of the sentence-initial modifier (in words) as a covariate. We pre-registered this covariate too but ultimately decided that it is not a sensible predictor because half of trials (complement-coercion and matched controls) lacked a sentence-initial modifier entirely, and the other half had modifier lengths of 3-5. In other words, modifier-length is highly correlated with trial type (multicollinearity).

¹⁶Our pre-registration stated that we would use random intercepts only, but it is standard to start with the maximal random-effects structure and reduce stepwise (which the lead author did not know at the time); hence, we ultimately opted for stepwise reduction.

¹⁷The R pseudo-code for, e.g., the critical+1 model is:
 $\text{logRT} \sim \text{trialtype} + \text{wordLength} + \text{prevlogRT} + \text{freq} + (1 + \text{trialtype} \mid \text{participant}) + (1 \mid \text{item})$.

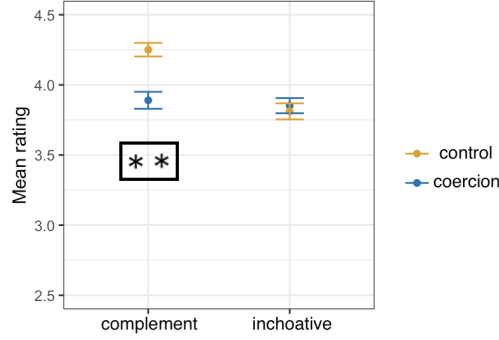


Figure 4: Mean ratings in Experiment 1a. Error bars are ± 1 SEM.

ordinal package (Christensen 2023). The models predicted rating from trial type, with by-participant random intercepts and slopes and by-item random intercepts. The first model used data from inchoative coercion trials, matched controls, and ungrammatical fillers; the second used data from complement coercion trials, matched controls, and ungrammatical fillers. The models used a Helmert coding scheme for “trial type” that allowed us to extract the following contrasts: ungrammatical vs. grammatical (where “grammatical” pools coercion and control) and coercion vs. control. For example, our first model compared (a) inchoative-coercion trials and controls vs. ungrammatical fillers; (b) inchoative-coercion vs. inchoative-control.

Our first model found that ungrammatical fillers were less natural than inchoative coercion trials+controls pooled together ($\beta = -1.77$, $SE = 0.32$, $p < 0.0001$), but ratings for inchoative-coercion trials did not significantly differ from inchoative-control trials ($\beta = -0.06$, $SE = 0.08$, $p = 0.4$). Our second model found that ungrammatical fillers were less natural than complement coercion trials+controls pooled ($\beta = -1.88$, $SE = 0.33$, $p < 0.0001$), and complement coercion trials were less natural than controls ($\beta = -0.48$, $SE = 0.09$, $p < 0.001$).

4.2.2 Experiment 1b

Figure 5 summarizes RT results for the analyzed subset of Experiment 1b’s data. The left panel displays mean logRTs for complement coercion trials like (c) in Table 2 and controls around the region of interest for complement coercion; the right panel displays logRTs for completive coercion trials like (a) in Table 2 and controls like (b) around the region of interest for completive coercion. Recall that Rett’s theory expects longer logRTs for trials like (a) vis-à-vis trials like (b) at verb or verb+1.

We analyzed the RT data using a pair of mixed-effects models (one fit on critical word logRTs, the other fit on critical+1 logRTs). The models had the same fixed-effects structure as the RT models in Experiment 1a; the random-effects structure was determined using the same procedure, with the final structure for

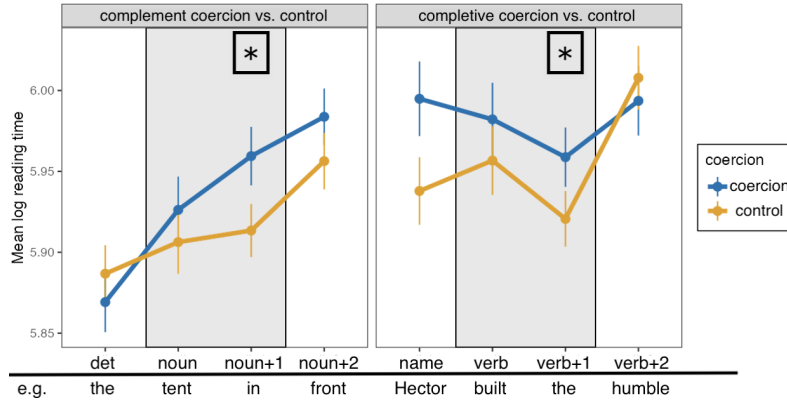


Figure 5: Mean logRTs in Experiment 1b. The regions of interest in each panel are highlighted in grey. Error bars are ± 1 SEM.

both models being by-participant and by-item random intercepts. As in Experiment 1a, we adopted a Bonferroni-corrected significance threshold of $\alpha = 0.025$ and custom-coded the trial type variable so as to extract the following three contrasts: (I) completive-coercion vs. completive-control; (II) complement-coercion vs. complement-control; (III) completive-coercion vs. complement-coercion.

The critical-word model found no significant effects. The critical+1 model found that completive coercion trials had longer logRTs than matched controls at verb+1 ($\beta = 0.033$, $SE = 0.013$, $p = 0.01$) and that complement coercion trials had longer logRTs than matched controls at noun+1 ($\beta = 0.035$, $SE = 0.013$, $p = 0.01$).

Figure 6 shows mean naturalness ratings for the four critical trial types; for reference, the mean rating for ungrammatical fillers was 1.83.

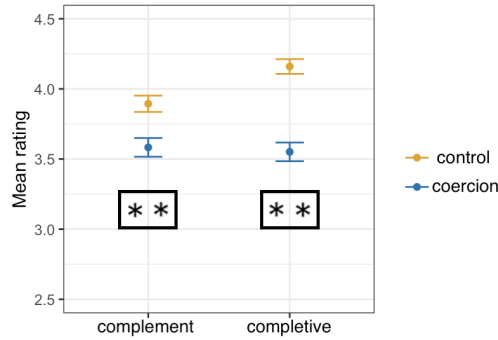


Figure 6: Mean ratings in Experiment 1b. Error bars are ± 1 SEM.

To analyze the naturalness data, we fit two mixed-effects ordinal logistic regressions in the same manner as Experiment 1a. The first model, which used data from completive coercion trials, matched controls, and ungrammatical fillers, found

that ungrammatical fillers were less natural than completive coercion + controls pooled ($\beta = -1.4$, $SE = 0.15$, $p < 0.0001$), and completive coercion trials were less natural than controls ($\beta = -0.53$, $SE = 0.08$, $p < 0.0001$). Our other model, which used data from complement coercion trials, matched controls, and ungrammatical fillers, found that ungrammatical fillers were less natural than complement coercion + controls pooled ($\beta = -1.29$, $SE = 0.19$, $p < 0.0001$), and complement-coercion trials were less natural than controls ($\beta = -0.25$, $SE = 0.07$, $p < 0.001$).

4.3 Discussion

Experiment 1a did not replicate Brennan & Pylkkänen’s (2010) findings about atelic sentences with *within*-modifiers: while the atelic *within*-modifier sentences in Brennan & Pylkkänen’s (2010) stimuli had slower reading times than controls in the region where the inchoative coercion account expects a slowdown, we found no such effect in Experiment 1a. This replication failure calls into question the claim that inchoative coercion is needed to account for atelic *within*-modifier sentences. Experiment 1a also did not replicate Brennan & Pylkkänen’s finding that atelic *within*-modifier sentences have lower offline ratings than controls. However, despite not finding an inchoative coercion effect, Experiment 1a replicated past work on complement coercion, suggesting that our null result on inchoative coercion cannot be chalked up to an overall inability to capture coercion effects using our paradigm.

While Experiment 1a did not find evidence for inchoative coercion, Experiment 1b found evidence for completive coercion: sentences with *at*-modifiers + accomplishments had slower logRTs than controls just after the main verb.¹⁸ Experiment 1b also found a complement coercion effect, serving as an additional sanity check.

¹⁸A reviewer worries about whether this is the correct takeaway from Experiment 1b based on the fact that, in Figure 5, we descriptively see a slowdown for completive coercion trials not just at verb+1, but also at “name.” Could the verb+1 effect in Experiment 1b be the result of inertia from an irrelevant cost earlier in the sentence? We think not. The first thing to note about the “name” effect is that it does not extend back into the adverbial; that is, if one visualizes logRTs for completive-coercion vs. completive-control throughout the whole sentence (i.e. beyond the window shown in Figure 5), we do not see a (descriptive) slowdown for completive coercion trials until “name.” This suggests that our effect is not driven by *at*-adverbials being harder to process than *in*-adverbials. Even so, the effect at “name” demands explanation; we can think of two possibilities. First, it could reflect an “anticipatory” coercion cost. In other words, perhaps some participants predicted the verb upon reading the name, leading them to coerce ahead of time. Second, perhaps the name was less predictable in completive-coercion trials vis-à-vis controls, triggering a surprisal-based slowdown unrelated to coercion. Evidence against the latter possibility comes from GPT-2 surprisal estimates (see Appendix for details on how these were extracted), which suggest that the name’s surprisal was actually lower on average in completive-coercion trials (mean surprisal = 3.67) vis-à-vis controls (mean surprisal = 4.31). That leaves us with the first possibility: that the “name” effect reflects an anticipatory coercion cost.

Finally, sentences with completive and complement coercion, like certain other coercion cases (see section 2.2), were less natural than controls in Experiment 1b.

At this point, the main question remaining about Experiment 1 concerns why the results of Experiment 1a conflicted with those of Brennan & Pylkkänen (2010). There are a few possibilities. First, while we only compared *within*-adverbial sentences to *for*-adverbial sentences, many of Brennan & Pylkkänen’s (2010) comparisons were less minimal. Brennan & Pylkkänen report an unexpected pre-verbal cost for their controls vis-à-vis their coercion sentences; hence, Brennan & Pylkkänen’s use of non-minimal comparisons may underlie the differences in our results. Second, our target sentences used *be*-statives, whereas theirs used subject experiencer verbs. The inchoative-coercion theory does not predict a processing asymmetry between these kinds of sentences, but it is nonetheless conceivable that self-paced reading is only sensitive enough to detect inchoative coercion’s effect when it occurs with subject-experiencers. A third possibility, raised by an anonymous reviewer, has to do with our use of contexts. In particular, it has previously been argued that embedding a coercion sentence in a context that establishes the relevant “event sense” attenuates the processing cost of coercion (Traxler et al. 2005, complement coercion), and the reviewer worries that our contexts laid out the full event sense for inchoative-coercion trials but not for complement-coercion trials.

We think that our results so far cannot adjudicate between the first two explanations; however, we consider the third explanation unlikely for two reasons. First, in Experiment 1b, we found a significant completive coercion effect despite using similar contexts as in Experiment 1a, and there is no reason to think that our contexts would nullify the cost of inchoative coercion and not completive coercion. Second, if we adopt Traxler et al.’s (2005) use of the term “event sense,” then our contexts in Experiment 1a established the event sense for complement-coercion trials as well as inchoative-coercion trials, suggesting that our contexts are unlikely to explain the difference between the two trial types. To see this, note that Traxler et al.’s (2005) finding is that while complement-coercion is costly in a discourse like (27a), it is not significantly costlier than controls in a discourse like (27).

- (27) a. The contractor had been building in the suburbs. That spring, he began a condominium next to the shopping center.
b. The contractor built a condominium in the spring. After he began the condominium next to the mall...

Our contexts are more akin to (27b) than (27a), in the sense that they mention not just the relevant type of event but also the noun used in the complement-coercion target sentence. For example, the contexts in (25) mention not just that Alex was reading but state more specifically that he read an article (“he read one of his own research articles...”). Thus, our contexts establish the “event sense”

(in Traxler et al.’s use of the term) for the complement-coercion target sentences. In this light, our findings on complement coercion can be thought of as a failure to conceptually replicate Traxler et al. (2005), which is independently interesting. Regardless, the main point is that one cannot appeal to Traxler et al. (2005) to explain why we replicated on complement coercion but not inchoative.

5 Experiment 2

Since Experiment 1b found a completive coercion effect in *at*-modifier sentences in self-paced reading, Rett (2020) might expect that completive coercion in *before*-clauses would have a similar self-paced reading cost. Experiment 2 tests this prediction, looking at the processing of telic *before*-clauses in two types of contexts: (a) contexts that are only compatible with a before-finish reading; (b) contexts that are compatible with a before-start reading and thus should not require coercion.

5.1 Methodology

5.1.1 Participants and materials

120 self-reported native English speakers from Prolific participated in Experiment 2, each of whom was paid £4.50 for their participation. Extreme RTs (defined as in Experiment 1) were excluded from analysis, along with data from 30 participants who gave a “True” rating on at least three of the six false fillers.

The structure of each trial in Experiment 2 was the same as in Experiment 1 (context, then target sentence, then rating). However, Experiment 2 had three critical trial types rather than four (see **Table 3**). On no-coercion trials like (a) in Table 3, participants read target sentences with accomplishments in *before*-clauses after seeing before-start contexts like Context A, in which Emma’s irritation emerged before Hector started building the tent. Completive coercion trials like (b) used the same target sentences as no-coercion trials but presented them alongside before-finish contexts like Context B, in which Emma’s irritation emerged after Hector started building the tent but before he finished. Finally, complement coercion trials like (c) presented complement-coercion target sentences with *finish* after contexts like Context B. As in Experiment 1, complement coercion trials were included as a sanity check; following past literature, we expect longer RTs for (c)’s target sentence vis-à-vis (a)’s target sentence at or just after *tent*.

- (28) a. **Context A:** Hector is a lifelong hiker and his daughter Emma wants to learn more about the outdoors, so they went on a hiking trip in the woods. It got very windy, so they decided to build a tent and wait in it until the wind stopped. Hector pulled out the camping equipment at

4pm and took longer than usual to set the tent up because of the wind; however, the tent was fully up by 4:30pm. Emma, who had become very irritable when the wind started (2pm), regained her composure once she saw the full tent at 4:30pm.

- b. **Context B:** Same first three sentences as Context A, then: Emma, who had become very irritable when Hector made a big mistake and had to start over at 4:15pm, regained her composure once she saw the full tent at 4:30pm.

Trial Type	Context	Target Sentence
(a) no-coercion	A	Emma was irritable before Hector built the tent in the woods.
(b) completive	B	Emma was irritable before Hector built the tent in the woods.
(c) complement	B	Emma was irritable before Hector finished the tent in the woods.

Table 3: Sample stimuli for Experiment 2. A sample trial is available at [this link](#).

Note that since Context A and Context B both involve at least partial overlap, Krifka (2010) does not predict a difference between completive and no-coercion trials (as established in section 3), in contrast to Rett (2020).

As participants were reading a context, they saw timelines to help them keep track of the events that would be mentioned in the upcoming target sentence (Figure 7). For example, for the trials in Table 3, a timeline showing the duration of Hector building the tent appeared once participants started to read the third sentence of Context A or B (Figure 7i). Upon moving on to the final sentence (i.e. the sentence about Emma’s irritation), the timeline morphed into either the one in Figure 7ii (if they were reading Context A) or the one in Figure 7iii (for Context B).

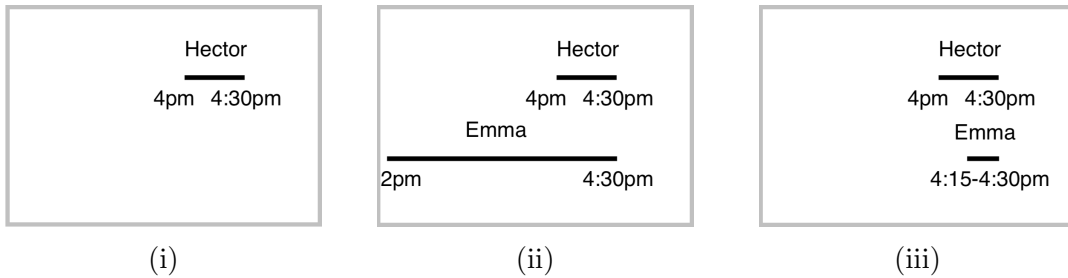


Figure 7: Timelines used on the trials in Table 3.

One other difference between Experiments 1 and 2 bears mention. In Experiment 1, participants rated the target sentence’s naturalness on every trial. In Experiment 2, participants gave a naturalness rating on critical trials and ungrammatical fillers but made a binary “True or False” judgment instead on grammatical fillers. We

made this choice because we hypothesized that binary judgments might be a better gauge of attention than Likert ratings (see fn. 11).¹⁹

5.1.2 Procedure

After an instruction text and a practice trial (accessible [here](#)), participants were sorted into one of four lists, each of which contained 12 critical trials (3 types x 4) and 12 fillers (4 true, 6 false, and 2 ungrammatical). The ungrammatical and true fillers were the same across lists, as were two of the false fillers. The other four false fillers in each list drew from the same pool of contexts as critical trials but used false *after*-sentences (e.g. Context A in Table 3 + *Emma was irritable after Hector built the tent in the woods*). For these fillers and the critical trials, we created 16 item sets like the item set in Table 3 (each centered around a different accomplishment) and used the Latin-square pattern detailed in section 4.1.2 to ensure that each participant saw one trial from each item set.

5.2 Results

The left panel of Figure 8 shows logRTs for complement coercion trials (e.g. (c) in Table 3) vs. no-coercion trials like (a) around the critical region for complement coercion; the right panel displays logRTs for completive coercion trials like (b) in Table 3 vs. no-coercion trials in the *before*-clause. Recall that under Rett’s theory, we predict a cost for completive trials vis-à-vis no-coercion, and we *prima facie* expect this slowdown to occur at the main verb in the *before*-clause or verb+1.

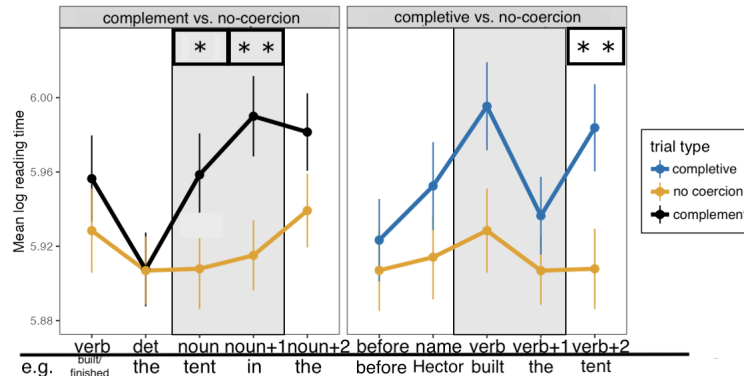


Figure 8: Mean logRTs in Experiment 2. Error bars are ± 1 SEM.

¹⁹Despite this hunch, the exclusion rate (30 out of 120) was similar to Experiment 1’s, suggesting that Experiment 1’s high exclusion rate was not due to its use of Likert-item attention checks but rather due to, e.g., participant fatigue. Whatever the cause of these high exclusion rates, we believe the results suggest that our manipulations were effective for attentive participants. In Experiment 3, we returned to Likert exclusions but used a slightly more lenient exclusion criterion.

To analyze the complement coercion results, we fit two mixed-effects models on the data from complement and no-coercion trials (one on direct object noun logRTs, one on noun+1 logRTs). The models predicted logRT from trial type (no-coercion vs. complement), with the same covariates as Experiment 1 and the random-effects structure being determined in the same way. Since we probed for the same effect in two regions, we adopted a Bonferroni-corrected threshold of $\alpha = 0.025$ for these models. Complement trials had longer logRTs than no-coercion trials at noun ($\beta = 0.058$, $SE = 0.019$, $p = 0.002$) and noun+1 ($\beta = 0.057$, $SE = 0.017$, $p = 0.001$).

Furthermore, for each of the critical words for completive coercion (verb, verb+1), we fit mixed models with the same structure as above on the data from completive and no-coercion trials. We found no significant logRT differences between the two trial types at verb ($\beta = 0.036$, $SE = 0.022$, $p = 0.11$) or verb+1 ($\beta = 0.008$, $SE = 0.018$, $p = 0.67$). Note that although the difference between no-coercion vs. completive at verb looks sizable in Figure 8, the verb model is telling us that this effect is not robust when controlling for factors like logRT on the previous word.

In addition to the above analyses, which tested our pre-registered hypotheses, we performed an exploratory analysis. In particular, since visual inspection of Figure 8 suggests a sizable slowdown for completive coercion trials (relative to no-coercion) at verb+2 (aka “noun”) and onward, we fit another model on verb+2 RTs for these two trial types. The model found that completive trials were read significantly more slowly than no-coercion trials at verb+2 ($\beta = 0.06$, $SE = 0.02$, $p = 0.003$). This effect is still significant even when we Bonferroni-correct for the fact that we probed for a completive coercion effect in three different regions ($\alpha = 0.05/3 = 0.016$).²⁰

Finally, Figure 9 shows mean naturalness ratings for the three critical trial types.

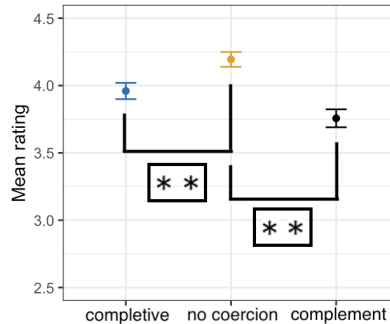


Figure 9: Mean naturalness ratings, Experiment 2. Error bars are ± 1 SEM.

²⁰As another exploratory analysis, we probed if there was within-experiment learning that led the completive coercion effect to become diluted. We found that if we fit a model on the subset of the data from the first no-coercion trial and completive trial that each participant saw, there is a marginal cost for completive trials at verb ($\beta = .101$, $SE = 0.05$, $p = 0.045$). Although $p < 0.05$, we opt not treat this effect as significant because the p-value is above a Bonferroni-corrected α .

For these data, we fit a mixed-effects ordinal logistic regression on the three critical trial types; it predicted rating from trial type (no-coercion as reference), with random intercepts and slopes for participants and items. Completive trials were less natural than no-coercion ($\beta = -0.72$, $SE = 0.23$, $p < 0.01$), and complement trials were also less natural than no-coercion ($\beta = -1$, $SE = 0.25$, $p < 0.01$).

5.3 Discussion

Experiment 2 replicated past findings on complement coercion, verifying that we can capture coercion effects using our paradigm and stimuli. Additionally, while Rett’s (2020) predicted slowdown in completive trials did occur, it started later than expected: at verb+2 rather than at verb or verb+1.

We can think of two ways for Rett (2020) to explain why the completive coercion effect in Experiment 2 was delayed to verb+2 (and why the complement coercion effect was not similarly delayed); the viability of either of these explanations needs to be assessed further, however. The first explanation concerns differences between the $n+1$ spillover word for completive coercion vs. complement coercion. The $n+1$ spillover word for completive coercion, i.e. verb+1, was always *the*. Not only is this word short and maximally frequent, but it was fully predictable in our stimuli: fillers and critical trials alike all had *the* in this position. Participants may have internalized the predictable nature of verb+1 as the experiment went on; as a result of the predictability, frequency, and short length of our verb+1, perhaps participants breezed through this word on many trials before the processing difficulty engendered by the verb had time to manifest behaviorally. If this explanation is right, any spillover effects from the verb should show up on verb+2 rather than verb+1.

Although the $n+1$ spillover word for complement coercion, noun+1, was always short and frequent as well, it was less predictable: the preposition or complementizer used as noun+1 was not the same across items. Thus, participants had less motivation to skim over noun+1 vis-à-vis verb+1. In sum, the idea is that for completive coercion, any spillover effects are likely to show up on verb+2, as verb+1 is too light and predictable to soak up processing load; however, a parser who skims over verb+1 might not skim over noun+1 (since noun+1 is less predictable), meaning that spillover effects for complement coercion are more likely to appear on noun+1.

While this is one type of explanation for the completive vs. complement asymmetry, it is not the only one. Another explanation relies on the fact that, as mentioned in section 3, completive coercion in *before*-clauses must be thought of as pragmatic (i.e. needed to repair a mismatch between a sentence and the context), whereas complement coercion is semantic (i.e. needed to repair a type mismatch). Perhaps the cost incurred by “pragmatic” coercions (whose processing has not been studied before now) is slightly delayed relative to “semantic” coercions because it takes longer to notice a mismatch between a sentence and a context than it does to

notice a type mismatch. On this view, the costs observed in Experiment 1b and 2 both reflect coercion, but the cost in *before*-clauses is delayed relative to *at*-modifier sentences because it takes longer to realize that coercion is necessary in the former than in the latter. This explanation dovetails psycholinguistic theories that posit a lag between semantic and pragmatic processing (Huang & Snedeker 2009, a.o.).

To sum up, Experiment 2 suggests that before-finish readings of telic *before*-clauses incur a cost in offline ratings and online processing, though the cost was delayed relative to predictions and thus needs to be probed further. Rett’s (2020) theory can explain this asymmetry between completive and no-coercion trials, in contrast to competitors such as Krifka (2010) (see section 3.2) and Condoravdi (2010) (see General Discussion); hence, Rett (2020) has a leg up when it comes to explaining Experiment 2’s results, even though some questions remain open for her.

6 Experiment 3

Having followed up on the results of Experiment 1b, we now address Experiment 1a’s failure to replicate Brennan & Pylkkänen (2010). Recall that this replication failure could have multiple sources, including (a) the use of non-minimal comparisons in Brennan & Pylkkänen (2010); (b) our use of *be*-statives rather than subject-experiencers. To adjudicate between these two possibilities, we repeated Experiment 1a with subject-experiencer verbs.

6.1 Methodology

As in Experiment 1a, we recruited 110 self-reported native English speakers from Prolific, each of whom was paid £4.50 for their participation. Extreme RTs (defined as above) were excluded, along with data from 14 participants who gave a rating of 3 or higher on at least 4 out of 6 false/ungrammatical fillers.

The design and procedure of Experiment 3 were identical to Experiment 1a except for the fact that Experiment 3 used subject-experiencer verbs where Experiment 1a used *be*-statives. This change necessitated some alterations to the contexts. As an example of Experiment 3’s stimuli, consider the item set in **Table 4**.

- (29) a. **Context A:** One semester, Jessica took a class taught by a professor, Alex, whose lectures were extremely difficult to understand. Jessica had heard that Alex was more coherent in office hours, so she scheduled a 20 minute appointment with him. Alex did nothing but read one of his own research articles to Jessica in full; Jessica voiced her annoyance about Alex’s behavior at first, but by the fifteen-minute mark, Jessica had given up and decided to tolerate Alex.

- b. **Context B:** Same first two sentences as Context A, then: Alex did nothing but read one of his own research articles to Jessica in full; Jessica tolerated Alex during the first fifteen minutes, but after that, she snapped and voiced her annoyance—all while Alex kept on reading.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Within fifteen minutes Jessica tolerated the unhelpful professor.
	(b) Control	B	For fifteen minutes Jessica tolerated the unhelpful professor.
Complement	(c) Coercion	A/B	Alex continued his article despite Jessica’s protest.
	(d) Control	A/B	Alex read his article despite Jessica’s protest.

Table 4: Sample stimuli for Experiment 3. A sample trial is available [here](#).

6.2 Results

Figure 10 summarizes RT results for complement-coercion trials vs. matched controls (left panel) and inchoative-coercion vs. matched controls (right panel).

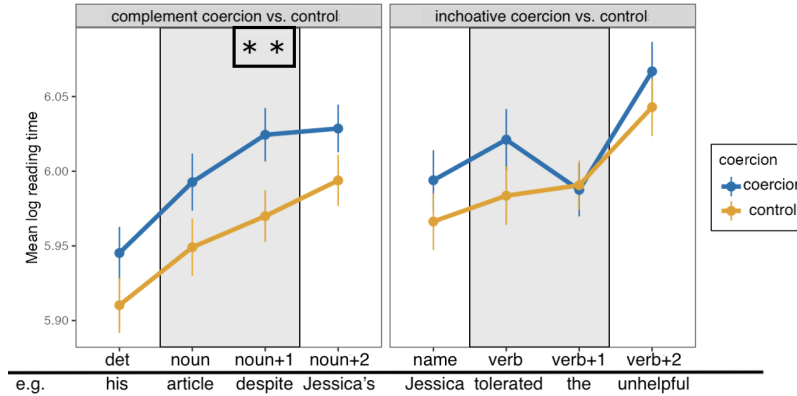


Figure 10: Mean logRTs in Experiment 3. Error bars are ± 1 SEM.

Recall that under the inchoative coercion theory of *within*-modifier sentences, we expect a slowdown for inchoative coercion trials at verb or verb+1. To ascertain whether this prediction was borne out (and to assess whether we replicated past findings on complement coercion), we analyzed the RT data in the same way as in Experiment 1a, fitting one model on critical word logRTs and one model on critical+1 logRTs. The critical-word model found no significant effects. The critical+1 model, by contrast, found that complement coercion sentences had significantly longer logRTs than matched controls ($\beta = 0.035$, $SE = 0.013$, $p = 0.009$), though

the difference between inchoative coercion sentences and matched controls was still insignificant ($\beta = -0.003$, $SE = 0.013$, $p = 0.78$).

Figure 11 shows mean naturalness ratings for the four critical trial types. For reference, the mean naturalness rating for the ungrammatical fillers was 1.58.

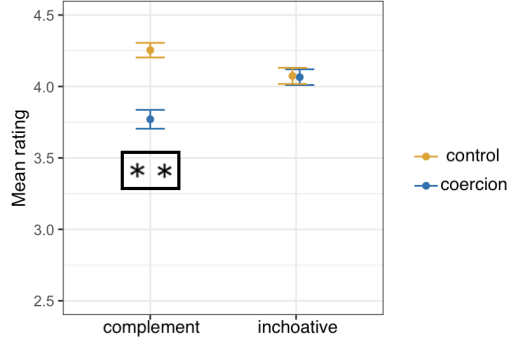


Figure 11: Mean naturalness ratings in Experiment 3. Error bars are ± 1 SEM.

We analyzed the ratings data as in Experiment 1a. Our first model found that while ungrammatical fillers were less natural than inchoative coercion trials + controls pooled together ($\beta = -1.8$, $SE = 0.26$, $p < 0.001$), ratings did not differ between inchoative coercion sentences and controls ($\beta = -0.035$, $SE = 0.07$, $p = 0.63$). In addition, ungrammatical fillers were less natural than complement coercion trials + controls pooled ($\beta = -0.6$, $SE = 0.1$, $p < 0.001$), and complement coercion trials were less natural than matched controls ($\beta = -1.74$, $SE = 0.3$, $p < 0.001$).

6.3 Discussion

Despite using the same types of predicates as Brennan & Pylkkänen (2010), Experiment 3—like Experiment 1a—failed to replicate Brennan & Pylkkänen’s findings about atelic *within*-modifier sentences. This suggests that the replication failure in Experiment 1a was not due to our predicate choice but some other factor, e.g. the use of non-minimal comparisons in Brennan & Pylkkänen (2010). Despite not finding a cost for inchoative coercion, Experiment 3 found an effect of complement coercion, suggesting that the results on inchoative coercion in Experiment 3 cannot be due to an overall inability to capture coercion effects with our contexts.

Whatever its source turns out to be, this continued replication failure calls a coercion-based theory of atelic *within*-modifier sentences into question.²¹ See

²¹Further complicating matters, recent work by Lisa Levinson and colleagues (accessible [here](#)) replicated Brennan & Pylkkänen (2010) with de-contextualized materials and the Maze task; thanks to a reviewer for flagging this. Clearly, more research is needed to adjudicate between a coercion-based theory of these sentences vis-à-vis an alternative like the theory in section 8.2.

section 8.2 for discussion of an alternative, non-coercive theory of these sentences.

7 Experiment 4

Regardless of the ultimate truth about atelic *within*-modifier sentences, it is still possible that there is coercion in atelic *after*-clauses where Rett (2020) claims there is. In other words, even if there turns out to be no inchoative coercion in atelic *within*-modifier sentences, we can still test the idea that after-start readings of *after* require coercion and after-finish readings do not. Experiment 4 tests this idea.

7.1 Methodology

7.1.1 Participants and materials

125 self-reported native English speakers from Prolific participated in Experiment 4, each of whom was paid £4.50 for their participation. Extreme RTs (defined as in section 4.2) were excluded, as were data from 40 participants who gave a rating of 3 or higher on at least four out of the eight ungrammatical/false fillers.

The structure of each trial in Experiment 4 was the same as in past experiments, and there were four critical trial types (see **Table 5**).²² On inchoative coercion trials like (a) in Table 4, participants read target sentences with subject-experiencer verbs in *after*-clauses alongside after-start contexts like Context A, in which Dave was regretful after Lara started fearing the dog but not after she stopped. Inchoative controls like (b) used the same target sentences as inchoative coercion trials but presented them alongside after-finish contexts like Context B, in which Dave was regretful even after Lara stopped fearing the dog. Note that since contexts A and B involve overlap to the same extent, Krifka (2010) does not predict a difference between inchoative coercion and inchoative control trials, in contrast to Rett (2020).

Unlike in Experiment 2, we could not implement a complement coercion sanity check by modifying the target sentences used in inchoative trials, as one cannot replace *after Lara feared the dog* with, e.g., *after Lara finished the dog* and preserve the subject-experiencer event sense. Instead, we implemented a complement coercion sanity check by comparing sentences like (c) and (d) in Table 4 in slightly different contexts like Context C. All complement coercion trials and matched controls used contexts without overlap between the matrix and embedded eventualities.

- (30) a. **Context A:** Lara, a physicist, was recently invited to present her work to Dave, a famous physicist and Nobel Prize winner, in a conference hall. Dave ended up bringing a large dog, and Lara is afraid of dogs.

²²Note that the items in Experiment 4 were similar to those in Experiments 1a and 3. We opt to show a different item here (Dave-and-Lara rather than Alex-and-Jessica) for a bit of variation.

While Lara felt fear towards the dog during the first fifteen minutes of her presentation (10-10:15am), she calmed down at the 15-minute mark and managed to immerse herself in her ideas. Dave had started to feel regretful about bringing the dog at 10:05, when he noticed how afraid Lara was; those feelings of regret went away once Lara regained her composure at 10:15.

- b. **Context B:** Same first three sentences as Context A, then: Dave had started to feel regretful about bringing the dog at 10:05, when he noticed how afraid Lara was; those feelings of regret lasted for many days, as he was sorry for having given Lara a few minutes of fear.
- c. **Context C:** Same first two sentences as Context A, then: So she smoked a cigarette outside the building (9:50-9:55am) to calm her nerves before the presentation. Dave passed by Lara at 9:56 and smelled the smoke; Dave hates smokers, so he immediately felt regretful about agreeing to meet Lara.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Dave was regretful after Lara feared the large dog.
	(b) Control	B	Dave was regretful after Lara feared the large dog.
Complement	(c) Coercion	C	Dave was regretful after Lara enjoyed a cigarette outside the building.
	(d) Control	C	Dave was regretful after Lara smoked a cigarette outside the building.

Table 5: Sample stimuli for Experiment 4. Click [here](#) to run through them yourself.

We used subject-experiencer verbs rather than *be*-statives simply because these were the predicates used by Brennan & Pylkkänen (2010); Experiments 1a and 3 suggest that this choice is likely not crucial. As in Experiment 2, participants saw timelines to help them keep track of events that would be mentioned in the target sentence. For example, on trials (a)-(b) in Table 5, participants saw a timeline showing the duration of Lara’s fear (Figure 12i) upon starting to read the third sentence of Context A or B. After moving on to the final sentence (i.e. the one about Dave’s regret), the timeline morphed into either the one in Figure 12ii (if they were reading Context A) or 12iii (if they were reading Context B).

7.1.2 Procedure

Participants saw the same instruction text and practice trial as Experiment 2 and were then sorted into one of four lists, each of which contained 16 critical trials (4 types x 4) and 10 fillers (2 true, 6 false, 2 ungrammatical). The fillers were the

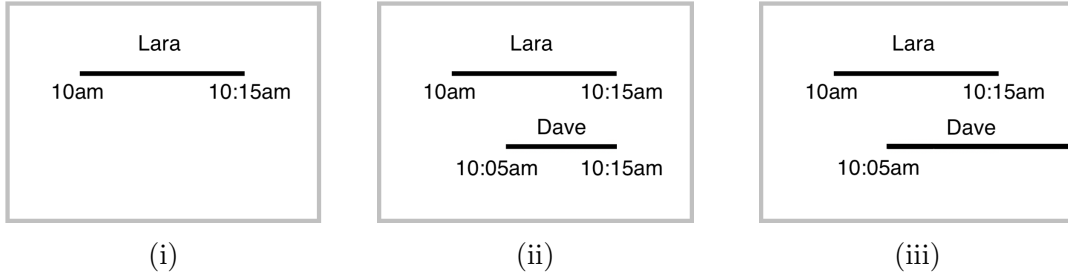


Figure 12: Timelines used on trials (a) and (b) in Table 4. Trials (c) and (d) used similar timelines for Dave’s regretfulness and Lara’s smoking.

same across the four lists; for the critical trials, we created 16 item sets like the one in Table 5 and used the Latin-square design from past experiments.

7.2 Results and analysis

The left panel of Figure 13 shows logRTs for complement coercion trials (e.g. (c) in Table 5) and matched controls like (d) in and around the region of interest for complement coercion; the right panel displays logRTs for inchoative coercion trials like (a) in Table 5 and controls like (b) in the *after*-clause.

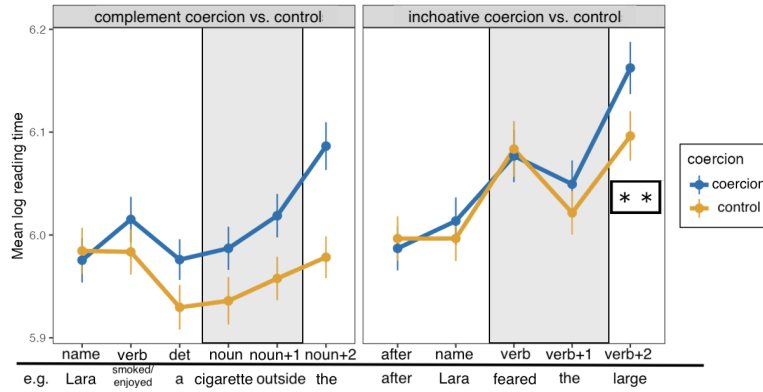


Figure 13: Mean logRTs in Experiment 4. Error bars are ± 1 SEM.

Starting with the left panel, we descriptively see a slowdown for complement coercion trials, but it seems to start earlier than expected (at “det”). To analyze these complement-coercion results, we fit two models on the data from complement and no-coercion trials: one on direct object noun logRTs, and one on noun+1 logRTs. The models predicted logRT from trial type (complement-coercion vs. complement-control), with the same covariates as in past experiments and the random-effects structure being determined in the same way. We adopted a significance threshold

of $\alpha = 0.025$ to correct for the fact that we probed for a complement coercion effect in two regions. The models did not find a significant effect at noun ($\beta = 0.02$, SE = 0.017, $p = 0.24$) or noun+1 ($\beta = 0.02$, SE = 0.023, $p = 0.35$). Note that since previous word logRT is a covariate, these results should be interpreted as suggesting that the complement coercion slowdown at noun is not significantly greater than the effect at the determiner; this dovetails with what we see in Figure 13.

Turning to the inchoative coercion results (Figure 13, right panel): for each of the *a priori* critical regions for inchoative coercion, we fit a mixed-effects model with the above structure on the data from inchoative coercion trials and matched controls. We did not find a significant difference between the two trial types at verb ($\beta = -0.014$, SE = 0.026, $p = 0.58$) or verb+1 ($\beta = 0.03$, SE = 0.019, $p = 0.12$).

In addition to the above analyses, which tested the hypotheses that were developed prior to any experimentation (see section 3), we performed the same sort of exploratory analysis as in Experiment 2, fitting a model on verb+2 logRTs for inchoative coercion trials vs. matched controls (same model structure as above). The model found that inchoative coercion trials were read significantly more slowly than matched controls at verb+2 ($\beta = 0.066$, SE = 0.024, $p = 0.008$). This effect is still significant even when we Bonferroni-correct for the fact that we probed for an inchoative coercion effect in three regions ($\alpha = 0.05/3 = 0.016$).

Finally, Figure 14 shows mean naturalness ratings for the four critical trial types. For reference, the mean rating for ungrammatical fillers was 2.27.

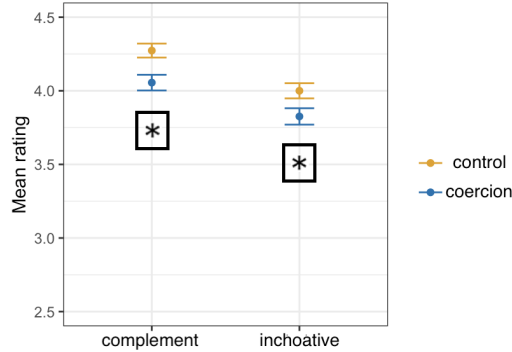


Figure 14: Mean naturalness ratings, Experiment 4.

We analyzed the ratings data in the same way as in Experiments 1a and 3. Ungrammatical fillers were less natural than inchoative coercion trials + controls pooled together ($\beta = -0.96$, SE = 0.13, $p < 0.001$), and inchoative coercion trials were less natural than controls ($\beta = -0.16$, SE = 0.07, $p = 0.01$). In addition, ungrammatical fillers were less natural than complement coercion trials+controls pooled together ($\beta = -1.09$, SE = 0.22, $p < 0.001$), and complement coercion trials were less natural than controls ($\beta = -0.16$, SE = 0.075, $p = 0.03$).

7.3 Discussion

Experiment 4’s results on *after*-clauses are akin to Experiment 2’s results on *before*-clauses. For one thing, inchoative coercion trials in Experiment 4 and complement trials in Experiment 2 were both less natural than controls, like some other cases of coercion. Furthermore, in both experiments, Rett’s (2020) predicted slowdown occurred but started later than expected: at verb+2 rather than verb or verb+1. The only way in which Experiment 4’s results differ from Experiment 2’s concerns complement coercion: while Experiment 2 replicated past findings on this coercion type, the complement coercion effect in Experiment 4 was diffuse and not localized to noun or noun+1; thus, our pre-registered analyses found no significant effect.

As in Experiment 2, there are two main ways for a proponent of Rett’s (2020) theory to explain why we saw an inchoative coercion effect at verb+2 in Experiment 4 rather than at verb or verb+1, though the feasibility of these explanations needs to be assessed further. The first explanation concerns the nature of verb+1 in Experiment 4. While verb+1 in Experiment 4 was not always *the* (cf. Experiment 2), it was always *the*, *a*, *his*, or *her*, i.e. a short, frequent, predictable determiner. As such, participants may have skimmed over verb+1 on many trials, leaving any leftover processing load from the verb to spill over onto verb+2.

Alternatively, one could interpret the later-than-expected slowdown in inchoative coercion trials as reflecting the fact that inchoative coercion in *after*-clauses is performed to resolve a different kind of mismatch than, e.g., complement coercion (“semantic” vs. “pragmatic” mismatches; see section 4.3 for more on this idea). Regardless of which of these explanations is correct, Experiment 4 suggests that after-start readings of atelic *after*-clauses are associated with an online cost and depressed naturalness ratings. Even though the timing of the cost raises additional questions for Rett (2020), she is nonetheless better-positioned than her competitors to explain the results of Experiment 4: her theory—unlike competitors such as Krifka 2010 (see section 3.2) and Condoravdi 2010 (see General Discussion)—predicts an asymmetry between inchoative-coercion and inchoative-control trials.

Regarding complement coercion, the messy results for this coercion type in Experiment 4 likely reflect an issue with its complement coercion stimuli rather than an issue with the experiment in general. To see this, consider (c)-(d) from Table 5. The embedded verb in (c) from Table 5, i.e. *enjoyed*, is likely less natural than the main verb in (d) given the context (Context C), since Dave’s regretfulness was caused not by Lara’s enjoyment of the cigarette but rather by the fact that she smoked it at all. This difference in the verb’s naturalness, which recurs in many of Experiment 4’s complement-coercion trials and controls, may have triggered a slowdown in its own right, leading to the sort of earlier-than-expected effect we observed. By contrast, the main verb in Experiment 2’s complement-coercion stimuli, which was always *finished*, is intuitively about as natural as the main verb in the

no-coercion stimuli; hence, the verb did not trigger a slowdown in that experiment.

8 General discussion

8.1 Main takeaways

This research tested a set of processing predictions that distinguish Rett’s (2020) theory of *before* and *after* from competing accounts. Rett argues that certain readings require completive or inchoative coercion, and therefore should exhibit the processing costs associated with coercion. Our results provide qualified support for Rett’s account. Experiment 1b found a processing cost in accomplishment *at*-modifier sentences, as predicted by the completive coercion analysis, and Experiment 2 found that *before*-clauses requiring completive coercion were harder to process than other *before*-clauses. Similarly, Experiment 4 found that *after*-clauses requiring inchoative coercion were harder to process than other *after*-clauses. In both Experiments 2 and 4, however, the processing difficulty emerged later than expected, requiring either the assumption that the delay reflects an experimental artifact or that pragmatically motivated coercion incurs a later cost than semantically motivated coercion. In addition, Experiments 1a and 3 failed to replicate earlier evidence that atelic *within*-modifier sentences involve coercion, weakening one of the motivations for extending a coercion-based analysis to *after*-clauses.

Despite these complications, the findings are more compatible with Rett’s account than with under-specification theories such as Krifka (2010). In particular, Experiments 2 and 4 revealed processing and rating asymmetries between alternative readings of *before*- and *after*-clauses that under-specification theories do not predict. More broadly, these experiments address an open question in the coercion literature. While coercion has traditionally been studied in cases involving semantic mismatches, Rett’s theory requires coercion in cases that are arguably pragmatic in nature. If our interpretation of the delayed effects in Experiments 2 and 4 is correct, pragmatically motivated coercion is indeed costly, but its processing signature emerges later than that of semantically motivated coercion.

8.2 A non-coercive theory of atelic *within*-sentences

As discussed above, the results of Experiments 1a and 3 chafe against inchoative-coercion theories of sentences like (31a) and thus raise the question of what a non-coercive theory of (31a) would look like. In this subsection, we sketch a non-coercive theory of this sort, building on intuitions from Krifka (2010).²³

²³Krifka (2010) pursues a non-coercive theory of (31a) because he is arguing against Condoravdi (2010), who (like Rett) posits an inchoative coercion mechanism; see section 8.3.

(31) Context: Ben was exposed to a virus at 6pm.

a. Within an hour, Ben was sick.

On the theory in question, *within_t* comes with an index *t*, which denotes the time-point that *within an hour* implicitly makes reference to. In context (31), *t* is 6pm (the time of Ben’s exposure). (32) shows the proposed truth-conditions of (31a) in this context, where *n* is the time of evaluation.

(32) $\llbracket \text{Within}_{6pm} \text{ an hour, Ben was sick} \rrbracket^n = 1$ iff
 $\exists t' \in [6pm, 7pm]: t' < n$ and Ben was sick at *t'*

Informally, (32) says that Ben was sick sometime between 6pm and 7pm (i.e. one hour afterwards). As it stands, these truth-conditions are too weak, since they do not entail that Ben started feeling sick after 6pm. To capture this additional meaning component, we assume (following Krifka 2010) that (31a) competes with alternatives where *an hour* is replaced by another time-span, e.g. (33a-b). The exact number of relevant alternatives depends on the context.

(33) a. Within forty-five minutes, Ben was sick.

b. Within half an hour, Ben was sick.

Within t', Ben was sick is stronger than *within t'', Ben was sick* whenever *t'* is a shorter time-span than *t''*. For example, in context (31), (33a) says that that Ben was sick sometime during [6pm,6:45pm], whereas (31a), as we have seen, says that Ben was sick sometime during [6pm,7pm]. As a result of this strength relationship, asserting (31a) implicates the negation of (33a) in a context where (33a) is relevant. Together, (31a) and the negation of (33a) entail that there was an event of Ben becoming sick during [6pm,7pm], mirroring intuitions.

Note that this theory derives the correct meaning for telic sentences like (34a) as well. On our theory, (34a) asserts that Hector ate the burrito in the interval [6pm,7pm] and implicates that he did not do so in a shorter interval.

(34) Context: Hector bought a burrito at 6pm.

a. Within an hour, Hector ate the burrito.

Although this analysis is only a sketch, it shows that one can account for *within*-modifiers without predicting that atelic *within*-modifiers will be uniquely hard to process. In particular, the above theory does not predict that atelic *within*-modifier sentences will be more costly than analogous *for*-modifier sentences, as processing both kinds of sentences involves computing similar implicatures (*within an hour* implicates, e.g., *not within thirty minutes*; *for two hours* implicates, e.g. *not for three hours*). Hence, the theory sketched in this subsection is more compatible with the results of Exps. 1a and 3 than an inchoative-coercion theory.

8.3 Condoravdi (2010)

As discussed in section 8.1, our results are more compatible with Rett’s (2020) ambiguity theory than with under-specification theories. However, there is an ambiguity theory of *after* other than Rett 2020 that we have not yet discussed: namely, that of Condoravdi (2010).

Condoravdi (2010), like Rett (2020), posits coercion; unlike Rett (2020), though, she claims that coercion in *before-/after*-clauses is performed to resolve a type mismatch. In particular, in a sentence like *Emma left after Hector was happy*, Condoravdi (2010) posits that *after* wants a time-interval as its first argument, whereas propositions like *Hector was happy* denote sets of time-intervals.

According to Condoravdi (2010), the type mismatch between *after* and its sister gets resolved via the insertion of one of two coercion operators. The first is EARLIEST, which is akin to Rett’s inchoative coercion mechanism. For example, if Hector was happy from 6pm to 8pm, then $\text{EARLIEST}(\llbracket \text{Hector be happy} \rrbracket) = 6\text{pm}$. Since 6pm is a (degenerate) time-interval, it can serve as *after*’s first argument.

In addition to EARLIEST, Condoravdi (2010) claims that an operator MAX can also coerce $\llbracket \text{Hector be happy} \rrbracket$ into a an interval, namely the largest interval throughout which Hector was happy. For example, if Hector was happy from 6pm-8pm, then $\text{MAX}(\llbracket \text{Hector be happy} \rrbracket) = [6\text{pm}, 8\text{pm}]$. Condoravdi’s MAX is not to be confused with Rett’s MAX_R . The latter always outputs a time-point, whereas the former outputs a larger interval; furthermore, Rett (2020) claims that MAX_R is always present in *before-/after*-sentences (regardless of the reading), while Condoravdi (2010) claims that MAX and EARLIEST compete for the same slot.

To summarize, in our example where Hector was happy from 6pm-8pm, *Emma was happy after Hector was happy* is predicted to have two readings, one in which EARLIEST coerces the *after*-clause and one in which MAX does. The LF with EARLIEST has an after-start meaning, and the LF with MAX has an after-finish meaning.

- (35) a. $\llbracket [\text{Emma be happy}] [\text{after} [\text{EARLIEST} [\text{Hector be happy}]]] \rrbracket = 1$ iff
 $\exists t' [\text{Emma is happy at } t' \text{ and } t' > 6\text{pm}]$
 b. $\llbracket [\text{Emma be happy}] [\text{after} [\text{MAX} [\text{Hector be happy}]]] \rrbracket = 1$ iff
 $\exists t' [\text{Emma is happy at } t' \text{ and } t' > [6\text{pm}, 8\text{pm}]]$

Since both readings of atelic *after*-clauses involve coercion for Condoravdi (2010), she does not straightforwardly predict any online or offline asymmetry between the two readings. To account for the results of Experiment 4, Condoravdi (2010) could try to say that coercion with EARLIEST is costlier than coercion with MAX, but this claim would require independent motivation (which does not exist, as far as we know). Note that even if coercion with EARLIEST can be independently shown to be costlier than coercion with MAX, Condoravdi’s timing predictions would not be different from Rett’s: parsers should know that the costlier coercion operator,

EARLIEST, is required at the main verb in the *after*-clause, so that is where we should see a slowdown. Thus, Condoravdi’s (2010) theory is at best on the same level as Rett’s theory when it comes to explaining the results of Experiment 4.²⁴

8.4 Broader implications and directions for future research

By providing moderate support for Rett’s idea that there are coercion mechanisms in *before/after*-clauses that are similar to overt morphemes in other languages, these results contribute to building a typology of aspectual coercion mechanisms. By extension, these results touch on the broader question of whether languages that lack a certain functional morpheme overtly have a coercion mechanism that does similar semantic work; this question arises not just in the domain of aspect but also domains like definiteness and kind-reference (Chierchia 1998). In addition to these implications for formal semantic theory, our results have consequences for theories of processing as well: in particular, if our explanation of the delayed costs in Experiments 2 and 4 is on the right track, then that would support the idea of a lag between semantic and pragmatic processing (Huang & Snedeker 2009, a.o.)

More broadly, if our interpretation of the results is correct, our findings underscore that meaning composition is very flexible, with repair mechanisms able to apply not just for semantic reasons but for pragmatic ones too. These mechanisms are not unconstrained, however, as shown by the fact that, e.g., completive coercion cannot apply to atelic predicates; if it could, then we would wrongly predict that *Emma was happy before Hector was happy* should have a before-finish reading. So our ability to repair sentences for pragmatic reasons, while powerful, is still constrained by grammatical factors like lexical aspect.

The most important direction for future research is to assess whether our results on *before/after* reflect a more general phenomenon. To that end, it would be useful to compare the processing of coercion in *before-/after*-clauses to the processing of other cases of “pragmatic coercion” (e.g. the habitual reading of *Mike swam for twenty years*, see section 2.2), in order to see whether the delayed effect observed in Experiments 2 and 4 is a signature of pragmatic coercion in general.

Another way to assess the generality of our results would be to probe whether they extend to similar ambiguities in other domains. For example, as mentioned in section 3.1, comparatives with an existential modal in the *than*-clause exhibit

²⁴Regarding telic *before*-clauses, Condoravdi (2010) claims that they have just one LF, the one with EARLIEST. LFs with MAX are, she claims, ill-formed when the temporal clause is telic (or non-cumulative more broadly, see p. 902). To capture the full range of readings, she claims that EARLIEST(*p*) picks out the telos when *p* is telic, not the start-point. Hence, her theory of *before* is essentially an under-specification theory (in the sense that telic *before*-clauses have just one LF, an LF with a before-finish meaning). In other words, she does not posit a distinction between the two readings of telic *before*-clauses, making the results of Experiment 2 difficult to explain for her.

a “polar asymmetry” that resembles the truth-conditional asymmetries between *before* and *after*, and Rett (2015) provides a theory of this “polar asymmetry” that has the same shape as Rett’s (2020) theory of *before/after*.²⁵ If the weak reading of these *than*-clauses, like the weak readings of *before/after*-clauses, incurs a cost relative to the strong reading, then we could conclude that the dynamics at play in *before-/after*-clauses are not a peculiarity of the temporal domain but rather recur in other ordered domains as well.

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²⁵In order to update Rett’s (2015) theory so as to perfectly line up with Rett 2020, all one would need to do is to say that MAX_R (9) is always present in both positive and negative comparatives (deriving the strong readings of (22a-b)), and there is an additional mechanism that can coerce the *than*-clause into its least informative point (deriving the weak reading of (22b)).

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Appendix

In this appendix, we repeat the reading-time analyses in which we found significant effects but include critical-word surprisal as a covariate, following the suggestion of a reviewer. Surprisals were extracted from the open-source large language model GPT-2 medium (Radford et al. 2019) using the *minicons* package in Python (Misra 2022). Each context + target sentence pair was treated as its own “text,” so surprisals for our target sentences took the preceding context into account.

Table 6 summarizes our findings; see Supplementary Materials for details.

Experiment	Coercion type	Mean critical-word surprisal (Coercion vs. control)	Outcome	β	SE	p
1a	Complement	4.04 vs. 2.26	Still sig.	0.037	0.017	0.02
1b	Completive	5.78 vs. 3.70	No longer sig.	0.025	0.015	0.09
1b	Complement	3.77 vs. 1.57	No longer sig.	0.027	0.015	0.07
2	Complement (noun)	3.34 vs. 0.973	No longer sig.	0.005	0.03	0.88
2	Complement (noun+1)	3.34 vs. 0.973	Still sig.	0.068	0.023	0.003
2	Completive	5.83 vs. 5.87	Still sig.	0.06	0.02	0.003
3	Complement	4.95 vs. 1.84	No longer sig.	0.016	0.015	0.29
4	Inchoative	8.13 vs. 8.04	Still sig.	0.067	0.025	0.008

Table 6: Results of the analyses with surprisal included as a covariate.

We see that most but not all of our complement coercion effects are now insignificant; this finding can be thought of as dovetailing with Delogu et al.’s (2017) conclusion that the complement coercion effect largely (but not entirely) stems from surprisal. The completive and inchoative coercion effects in Experiments 2 and 4 are still significant, but the completive coercion effect in Experiment 1b is not.

Should we conclude that Experiment 1b yields no evidence for completive coercion in *at*-modifier sentences after all? We think not. First, we could be dealing with a “causal bottleneck” (Levy 2008), as a reviewer notes, where the relationship between coercion and processing is mediated by surprisal. In other words, even if Rett’s theory of accomplishment *at*-modifier sentences is correct, it is conceivable the need for repair in these sentences causes them to be used less often, which in turn causes the verb in such sentences to have higher surprisal than controls.

Second, we think that attributing the effect in Experiment 1b to surprisal, full stop, would not be explanatory. Table 6 indicates that the completive effect in Experiment 1b is driven by accomplishments being more probable after *in*-adverbials than after *at*-adverbials, but that just pushes the question a level higher: why would accomplishments be more probable with one type of adverbial? The only answer we can think of is a restatement of the completive coercion theory: accomplishments are less probable with *at*-adverbials because there is a semantic conflict between the durative verb and the punctual adverbial.